



CEO compensation: Evidence from the field

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ABSTRACT

We survey directors and investors on the objectives, constraints, and determinants of CEO pay. We find that directors face constraints beyond participation and incentives, and that pay matters not to finance consumption but to address CEOs' fairness concerns. 67% of directors would sacrifice shareholder value to avoid controversy, leading to lower levels and one-size-fits-all structures. Shareholders are the main source of constraints, suggesting directors and investors disagree on how to maximize value. Intrinsic motivation and reputation are seen as stronger motivators than incentive pay. Even with strong portfolio incentives, flow pay responds to performance to fairly recognize the CEO's contribution.

1. Introduction

How to design executive pay is a first-order decision for every firm. Pay is critical for attracting, retaining, and motivating a CEO, and affects the wider company beyond the CEO – high pay may demotivate employees or damage a company's customer reputation. Even more broadly, CEO pay across the economy influences the public's perception of capitalism. As a consequence, CEO pay receives more attention than nearly every other routine corporate decision. Boards have dedicated

remuneration committees, investors have a special “say-on-pay” vote, and pay is highly regulated.

Due to its importance, it is critical to understand how CEO pay is set. Academics typically study this question through theorists building models and empiricists testing the models' predictions. While a great deal has been learned through these methods, they have limitations. First, data only documents the outcome of an optimization problem and not the underlying program that led to it. Even if data is consistent with a model, a very different model may have generated it; thus, even if a

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model claims empirical support, it may not describe the actual economic mechanism. Second, many key ingredients of compensation models, such as risk aversion, are difficult to measure and thus test.

This paper anonymously surveys non-executive directors and institutional investors on how they set or influence pay. Our first goal is to understand the *objectives* and *constraints* of contract design, and thus the underlying program. Our second goal is to identify the *determinants* of pay, by investigating factors that theories deem relevant but are unobservable, or that prior models have overlooked. We distributed the survey in late 2020, receiving responses from 203 non-executive directors of FTSE All-Share companies and 159 institutional investors in UK equities. Most questions allowed respondents to add free-text comments, and we conducted 14 post-survey interviews. The answers reveal several interesting results, organized into four groups, that point towards a more complex model of CEO pay than currently used in the academic literature:

1.1. Objective and constraints

Our first question asks respondents to rank the importance of three goals when setting CEO pay. Only 1% of directors and 5% of investors view keeping the level of pay down as their primary goal. This contrasts with the frequent opposition to high pay from policymakers, the public, and some investors, often irrespective of the implications for CEO retention and motivation. It is, however, consistent with CEO pay being a small percentage of firm value, while hiring a subpar CEO or providing suboptimal incentives has potentially large effects. 65% of directors view attracting the right CEO as most critical, while 34% prioritize designing a structure that motivates the CEO. For investors, these figures are 44% and 51% respectively. This reversal reflects a theme that recurs throughout our survey – directors view labor market forces, and thus the participation constraint, as more important than investors, who prioritize the incentive constraint.

However, boards feel restricted by far more than the participation and incentive constraints focused on by standard models. 67% of directors admit that they are willing to sacrifice shareholder value to avoid controversy on CEO pay – from parties such as proxy advisors, employees, and customers. Surprisingly, the strongest constraint is the need to obtain investor support, even though this should be automatic if boards are setting pay optimally. Instead, directors believe that shareholder guidelines, paradoxically, harm shareholder value. 77% report that such constraints have forced them to offer a lower level of pay, and 72% an inferior structure.

Most models of CEO pay take the “shareholder value” view that pay is set by a single principal, a shareholder-aligned board. The main alternative is the “rent extraction” view, whereby boards are captured by CEOs and thus do not seek to maximize value. However, our free-text fields and interviews suggest a third perspective – directors and investors share the same objective (shareholder value), but view the world differently. One possibility is that directors better understand the CEO labor market, whereas shareholders push for changes that would violate the CEO’s participation constraint or demotivate her (“uninformed investors”). Another is that boards overestimate the value of their CEO or underestimate their latitude to improve pay (“uninformed boards”).

To help disentangle these interpretations, we ask the 77% of directors who were forced to offer lower pay about the consequences. While 7% report that the CEO left, and 13% that they hired a less expensive CEO, 41% admit that there were no adverse effects. This result is meaningful, since any self-serving bias would discourage this response. Thus, at least in some cases, boards underestimated their latitude to cut pay. However, 42% report that the CEO was less motivated, suggesting that the level of pay affects incentives, in contrast to standard theories.

1.2. Incentives and variable pay

There is greater agreement on the second set of questions – the role of financial incentives in motivating CEOs. Both boards and shareholders believe they are relevant but of secondary importance. The CEO’s intrinsic motivation and personal reputation are seen as most important, yet are absent from nearly all theories. Notably, directors see CEO career concerns, the focus of a large literature, as third-order: fewer than 20% view the risk of being fired or a move to a larger firm as important motivators.

Both boards and investors believe that motivating the CEO is the main reason for offering variable pay, despite viewing financial incentives as secondary. The free-text responses and interviews suggest that financial incentives reinforce intrinsic and reputational incentives. CEOs believe it is fair to be rewarded financially for good performance; perceived unfairness would erode their intrinsic motivation. As one respondent stressed, “the retrospective acknowledgement of exceptional performance is important.” An increase in realized pay also signals the CEO’s performance to outsiders, boosting her reputation.

These responses suggest that incentive pay may work through different channels to standard models. In these models, the CEO improves firm value only if her utility from consuming the resulting pay increment exceeds the effort required – the contract offers sufficient *consumption incentives*. Our respondents instead suggest that a CEO does not need the extra pay to finance consumption, but believes it is fair to be recognized for a job well done – variable pay provides *ex-post recognition*.

The importance of ex-post recognition has two implications. First, it suggests that a CEO assesses her pay not only for the consumption utility it provides, but also against what she deems to be a fair reward. This fair reward is believed to be affected by at least two reference points – the CEO’s contribution to the company and the pay of her peers.

Second, flow pay plays a special role not provided by portfolio incentives. In standard theories, only total incentives matter – it is irrelevant whether they stem from changes in flow pay or the value of equity holdings. Empirically, incentives from equity holdings are much greater, so standard measures of CEO incentives ignore flow pay. However, changes in flow pay provide greater ex-post recognition, because they require a discretionary decision by the board and are voted on by shareholders. They are also publicly disclosed, boosting the CEO’s reputation. Thus, they may be important even if the CEO holds significant equity.

Directors and investors strongly support two other reasons for variable pay: attracting or retaining a high-ability or hard-working CEO, and for the CEO to share external risks with investors and stakeholders. The second reason is surprising and contradicts standard theories, since it implies inefficient risk-sharing. It is, however, consistent with a fairness model in which directors and investors also evaluate CEO pay relative to reference points, which for them include shareholder returns.

When asked about the split between fixed and variable pay, neither directors nor investors assign much relevance to CEO risk aversion and firm risk, the split’s main determinants in standard models. This indicates a wide gulf between the models and real-world practice.

1.3. Pay levels

Our third set of results concerns the level of pay. We first ask what determines the pay of a new CEO. Both directors and investors view CEO ability as most important. Unexpectedly, pay at peer firms is more important than the new CEO’s actual outside options, such as pay at her prior firm and at other firms she could move to. One explanation, supported by the free-text fields and interviews, is that peer pay matters because it is a reference point she uses to assess whether her pay is fair.

When asked about increases in expected pay for incumbent CEOs, both directors and investors state that the primary justification is good recent performance. This is surprising given the substantial equity

holdings that CEOs have, but is consistent with changes in flow pay providing ex-post recognition. Other justifications receive only minor support.

Directors also report that decreases in expected pay are rare, with 77% never having implemented one. Interestingly, while directors believe that good recent performance justifies pay increases, they do not believe that poor performance justifies decreases. Several interviewees explained that pay cuts would demotivate the CEO and, if performance were poor enough to warrant a cut, they would fire her instead. The two justifications for pay decreases with the most support are external pressure and financial constraints, likely because they are less detrimental to the CEO's sense of worth.

1.4. "Suboptimal" pay practices

Finally, we study the reasons for apparently suboptimal pay practices. The first is the limited use of relative performance evaluation. Directors support three explanations, all absent from existing theories. One is again fairness – CEOs should benefit from an upswing since investors and stakeholders do. The other two reasons are practical – it can be difficult to find suitable peers for some firms, or to observe peer performance for some performance measures. Explanations proposed by existing models, such as keeping pay competitive with peers during upswings, receive little support.

A second apparently suboptimal practice is the short-term nature of many pay incentives. Here, directors' and investors' views differ sharply. 78% of investors believe the CEO would make better decisions if incentives were more long-term. Fewer than 6% agree with each of three potential concerns – that long-term incentives are less effective motivators, would jeopardize CEO retention or recruitment, or would require a costly adjustment in pay level. In contrast, directors view incentives as already sufficiently long-term, and only 21% believe that further lengthening would improve decisions.

1.5. Relation to the literature

This paper builds on the literature on CEO compensation, reviewed by Murphy (2013), Edmans and Gabaix (2016), and Edmans et al. (2017). It is also related to other corporate finance surveys, such as Graham and Harvey (2001), Graham et al. (2005), Brav et al. (2005), Ben-David et al. (2013), and Graham et al. (2013). A unique feature is that we survey both directors and investors. The above papers study executives only; others, such as McCahery et al. (2016), Krueger et al. (2020), and Gompers et al. (2020), study investors only. CEO pay is arguably the corporate decision on which investors have the greatest influence, given say-on-pay, and thus an interesting setting to compare their views with directors.

This paper proceeds as follows. Section 2 discusses the motivation for the survey, how we conducted it, and the context of UK executive pay. Section 3 presents the results on the objectives and constraints of CEO pay design, Section 4 addresses the level of pay, and Section 5 its structure. Section 6 discusses generalizability and Section 7 concludes. The survey also included two questions on the influence of different parties in the pay setting process, which we discuss in Online Appendix A.

2. Motivation, method and context

2.1. Motivation: Standard theory vs. real-world practice

A key goal of the survey is to guide future compensation theories. Nearly all models contain three basic elements: an objective function that is firm value net of CEO pay, a participation or individual rationality ("IR") constraint, and incentive compatibility ("IC") constraints. These elements are so standard that they can be written down without any justification. However, they may not capture how practitioners actually set CEO pay, hence the value of a survey.

Starting with the objective function, it assumes that the only downside of high CEO pay is that it directly reduces shareholder value. However, it may also demotivate employees or harm the firm's reputation. Moreover, the board may have a different objective, such as winning a say-on-pay vote. Turning to the IR, it binds in many models. This implies that, if a CEO's pay were reduced by \$1, she would leave, which seems unlikely in practice. The IC may also not bind, if intrinsic motivation or reputational concerns are sufficient to motivate the CEO.

Considering the whole program together, it ignores many other potential constraints. Boards and CEOs do not negotiate pay in a vacuum. Norms on what investors and stakeholders deem acceptable may limit the level of pay or constrain its structure. The CEO's preferences, reflected in the IR and IC, are likely to be richer than a simple function of effort and consumption. CEOs may, for example, also care about their reputation, fairness, and being appreciated by directors and investors.

2.2. Surveys and archival research

The standard empirical methodology is archival research. This has several advantages, such as large datasets, objectivity, and the ability to control for multiple factors, but also limitations. First, it is difficult to measure key theoretical determinants of CEO pay, such as the CEO's risk aversion and cost of effort. Even if there seems to be a natural empirical measure, it may not be the only, or even the main one that practitioners consider. For example, shareholder returns are the main metric for CEO performance in empirical studies. However, the board and investors may also study strategic execution or other non-financial factors that are difficult for empiricists to observe, such as corporate culture.¹

Second, even if an empirical proxy can be found, a statistical relationship could have multiple causal and non-causal interpretations. For example, several studies have shown that CEO pay is correlated with peer pay.² This may be because boards are indeed benchmarking³, or because similar firms solve similar optimization problems without paying attention to peers. Empirical researchers are aware of these challenges and have attempted to use instruments or natural experiments; however, clean identification strategies are relatively rare in the setting of executive pay.⁴

Third, surveys allow us to investigate the *relative* importance of various pay determinants. For example, prior papers have individually studied the influence of proxy advisors, the media, and compensation consultants on CEO pay; we include all of them in the same question. Even if archival research includes several factors in the same regression, the one with least measurement error may appear to be most significant.

Finally, archival studies may be limited by the "academic paradigm," i.e., restricted to what existing research suggests is relevant. A survey that includes free text fields and is accompanied by interviews can uncover new determinants of CEO pay that have not previously been documented.

The survey methodology itself has limitations, which we have endeavored to attenuate. First, similar to archival research, our questions may be limited by the "academic paradigm." We addressed this

¹ Moreover, while academic researchers typically filter out peer performance, practitioners may also remove performance attributable to the CEO's predecessor or the company's pre-existing assets, such as its brand.

² See, e.g., Bizjak et al. (2008) and Cremers and Grinstein (2014).

³ Benchmarking in turn could have multiple causal explanations, such as learning from peers, addressing the CEO's outside option, or making it easier to justify a pay practice to investors.

⁴ As Edmans et al. (2017) write in their survey: "Given the nature of the pay-setting process, there are very few valid instruments that affect pay without also affecting the outcome variable of interest through some other channel (thus violating the exclusion restriction) ... measuring the causal effects of pay on behavior and performance remains one of the most important challenges of this literature and an open question for future research." Very similar challenges apply to identifying the causal determinants of pay.

concern by having a practitioner coauthor, beta-testing the survey to see if we had omitted important factors, and including free-text fields – if they systematically suggested a response outside our set of options, we explored it in interviews.

Second, respondents may interpret questions differently to how we envisaged. Our practitioner coauthor helped ensure that we translated academic concepts into language that respondents would be familiar with; in addition, we engaged in extensive beta-testing of the survey and provide free-text fields to detect any persistent misinterpretations.

Third, Friedman's (1953) "as if" critique is that contract designers may act in accordance with a theory but be unable to articulate it. We thus did not test theories whose mechanisms are so complex that respondents may not be conscious of them. Similarly, we caveat that low scores for an answer need not mean that it is immaterial for CEO pay, but only that respondents may be unaware of its relevance.

Fourth, respondents may also choose a response simply because it sounds logical. We reduce this risk by not providing rationales. For example, when exploring the determinants of incentives, we offer "how risky the firm is" rather than "the riskier the firm is, the weaker incentives are as they expose the CEO to too much risk." While the latter might identify the mechanism more precisely, a respondent might score it highly because it sounds sensible.

Finally, respondents may prefer socially acceptable answers. In addition to guaranteeing anonymity, we tried to not ask questions that would likely lead to misreporting. For example, we did not ask directors why pay is so high, as this may prompt rationalizations even if pay were unjustified, but instead asked what would happen if pay were reduced.

2.3. The UK context and COVID-19

We study executive pay in the UK for a number of reasons. First, the UK is between the US and continental Europe in both the level and structure of pay, suggesting that it is not an outlier. While pay levels are higher in the US and lower in Europe, this can mostly be explained by firms being larger and using more incentive compensation; indeed, these differences become moderate after adjusting for firm size and risk (Conyon et al., 2011; Fernandes et al., 2013). The differences in the structure of pay are illustrated in Table 1: in 2019, long-term incentives comprise 48% of total CEO pay in large UK firms, compared to 34% and 35% in Germany and France, and 71% in the US.⁵

Second, the UK does not have any differential tax treatments that would cause a preference for one form of pay over the other (unlike, for example, Section 162(m) in the US which, before 2018, exempted

Table 1

Pay mix and levels, UK, US, Germany, and France in 2019.

	Median CEO pay % of total (\$m)			
	UK	US	Germany	France
Salary	24% (1.30)	10% (1.28)	25% (1.68)	28% (1.35)
Bonus	29% (1.58)	19% (2.54)	41% (2.77)	36% (1.74)
Long-term incentives	48% (2.64)	71% (9.54)	34% (2.28)	35% (1.69)
Total	100% (5.52)	100% (13.36)	100% (6.73)	100% (4.78)

Source: CEO Pay Landscape in Japan, US, and Europe – 2019 Analysis, WillisTowersWatson. Firms with >\$10bn sales.

⁵ This comparison exaggerates the US-UK differences, because US bonuses are paid entirely in cash, while in the UK, typically half of the bonus is paid in deferred shares. If these were counted as long-term incentives, the proportion of UK CEO pay attributable to long-term incentives would exceed 60%, well above typical European levels (where deferred bonuses are less common) and closer to the US.

qualified performance-based pay from the \$1 million cap on tax deductibility), nor do accounting or disclosure rules have material asymmetric impacts. Thus, any findings are unlikely to be due to tax, accounting, or disclosure factors that are specific to the UK.

Third, the UK has a relatively settled regulatory regime surrounding executive pay, and UK investors have significant experience in evaluating, voting on, and engaging on CEO pay. The UK was the first country to introduce say-on-pay in 2003, which inspired similar requirements in the US and Europe. There has also been growing convergence across the globe in other regulations, disclosure requirements, and the behavior of investors and proxy advisors, with major investors voting and engaging in an increasingly consistent manner in both the US and Europe. Moreover, a number of pay practices, such as performance-based long-term incentive awards and relative performance evaluation, first became widespread in the UK before becoming prevalent elsewhere.

Fourth, on a practical level, the authors are from well-known UK universities, increasing the likelihood that UK directors and investors would respond to an academic survey. Similarly, the authors' names are known to many UK directors and investors, also increasing the response rate.

We believe that most of our conclusions will generalize to other countries. Not only are UK pay levels, structures, and disclosures not an outlier, but the UK regulatory regime has been mimicked elsewhere. That being said, there remain differences between the UK and other countries; for example, options are used far less than in the US. We thus did not ask questions on topics where the UK may be an outlier (e.g., the mix of stock vs. options). Section 6 discusses in more detail which results are likely to generalize to other countries and which may not.

Two areas where the UK differs from the US (but not necessarily the EU) are the say-on-pay regime and governance code. Every three years, a UK board's Remuneration Committee (RemCo) submits its remuneration policy to a binding "policy vote." This sets the framework within which the RemCo must make pay decisions. Every year, these decisions are put to a non-binding "implementation vote." The EU Shareholder Rights Directive has adopted this regime since 2014. In contrast, US firms only hold a non-binding implementation vote at least once every three years. UK RemCos are also constrained by the 2018 revision of the UK Corporate Governance Code (the "Code"); violations can lead to significant voting opposition. For example, the revised Code requires pension contributions for executive directors to be aligned with the wider workforce. Before 2018, CEO pensions had contribution rates of up to 50% of salary, while employee schemes were below 10%, so the Code led to large reductions for CEOs.

The survey was carried out during the COVID-19 pandemic, in which many firms reduced CEO pay. COVID-19 and the recent pension reductions enable us to study pay cuts, which are otherwise rare. To reduce the risk that responses were skewed by COVID-19, none of our questions focused on recent events but on pay practices in general, e.g., "Do you filter out industry conditions from all performance measures?" rather than "Did you filter out the impact of COVID?" Section 6 discusses the extent to which our results may have been influenced by the timing of the survey.

Online Appendix B provides additional context plus stylized facts on UK CEO pay.

2.4. Survey design and delivery

We benefited from extensive feedback on our questions before launching the survey. We presented the questions to academic audiences and sent them to leading researchers. We sent several drafts to the UK government's department of Business, Energy, and Industrial Strategy, which has significant experience in administering surveys. We beta-tested the survey with directors, investors, and compensation consultants to ensure that they were interpreting the questions as we intended, that the survey was not too long (our target was 15 minutes), and that we were not missing key dimensions. Most of these beta tests occurred

via Zoom, where the practitioners answered the questions aloud, so we could observe how they were interpreting them.

We launched the survey in November and December 2020. To encourage responses, we donated £100 for each completed survey (up to a total of £25,000) to the UK's National Health Service COVID appeal, and offered respondents the option to receive a draft of the working paper before its public release.⁶ We administered the survey using the Qualtrics online platform, giving respondents a generic (rather than individualized) link to guarantee their anonymity. Except for the demographic questions, we randomized the order of responses within each question.

We surveyed two types of respondent. The first was non-executive directors of FTSE All-Share companies (excluding investment trusts), identified from CapitalIQ and BoardEx; we attempted to contact every director. This contrasts with other approaches that involve handing out surveys at conferences or using alumni networks, which could lead to selection biases. The second was investors in UK equities, both asset managers and institutional asset owners. Within an asset manager/owner, there are fund managers and governance specialists (otherwise known as “stewardship” or “responsible investment” specialists). The latter coordinate voting and engagement across the investor's funds.

We attempted to contact every fund manager of all UK equity funds listed on Trustnet, a comprehensive database of funds offered in the UK.⁷ We also emailed the main contact for responsible investing at all signatories to the UK Stewardship Code⁸ and at members of the UK Sustainable Investment and Finance Association. For most, the listed contact was a stewardship specialist such as a Head of Responsible Investment; for others, it was a Chief Investment Officer (“CIO”), fund manager, or analyst. Online Appendix C provides further details on our distribution procedure.

In total, we contacted 1,312 non-executive directors of 421 firms and 556 investors at 231 asset managers or asset owners. We obtained responses from 203 directors and 159 investors; 170 (132) answered every question. This corresponds to a response rate of 15.5% for directors (13.0% who answered every question) and 28.6% (23.7%) for investors.⁹ This compares favorably with response rates of 4–15% for the survey papers referenced in Section 1. We then interviewed 14 respondents to explore the reasons behind their responses. The interviewees were selected because they filled in several free text responses and to create a mix of investors (asset managers and asset owners, and fund managers and governance specialists) and directors across industries.

Table OA1 presents summary statistics on the director respondents. 39% were from the FTSE 100, 36% from the FTSE 250, and 22% from the FTSE Small Cap index. 27% were board chairs, 33% RemCo chairs, and 24% other RemCo members. Table OA2 shows that the industry breakdown is representative of the FTSE All-Share index. The responses were similar across subsamples, including sector, size of company, and director role. We therefore tabulate aggregate results and highlight in the text cases where there were variations across subsets.

Table OA3 contains summary statistics for the investor respondents. 80% were from asset managers, 8% from asset owners, and 13% from

hybrids, such as pension funds. 52% were governance specialists, 26% fund managers, 8% stock analysts, and 6% CIOs. Where there is no confusion, we will use “fund managers” to encompass these last three categories. Fund managers are primarily evaluated according to returns, while governance specialists may be more sensitive to the societal consequences of pay. However, the responses from both groups were generally similar, as were responses from investors of different sizes. We will highlight the few cases where they differed.

3. The objectives and constraints of CEO pay design

3.1. Objectives

Our first set of questions aims to study directors' and investors' objectives and the constraints they are operating under. The standard CEO pay model described in Section 2.1 involves three objectives, assuming a given effort level: (1) minimizing the level of pay subject to (2) the structure inducing the required effort level (i.e., the IC being satisfied) and (3) the CEO accepting the contract (i.e., the IR being satisfied). While it is widely accepted that these are the three primary goals of a contract, practitioners' views of their relative importance have not been studied. Accordingly, our first question asks respondents to rank them (“Rank the importance of the following goals when setting CEO pay”).

Table 2 illustrates the results. Both directors and investors view minimizing the level of pay (“keep the quantum of pay down”) as least important, with 91% of directors and 85% of investors ranking it last.¹⁰ This is surprising given the frequent pushback against high pay from policymakers, journalists, the public, employees, and investors, often irrespective of its implications for CEO retention and motivation.¹¹ It is, however, consistent with “shareholder value” models where the CEO's talent (Gabaix and Landier, 2008) and effort (Edmans and Gabaix, 2011) have multiplicative effects on firm value, making attracting and motivating a talented CEO highly valuable, especially for large firms. Moreover, the low relative importance of pay restraint does not mean that it is irrelevant in absolute terms; indeed, future questions will show that both directors and investors view the quantum of pay as important, mostly because of external constraints.

Table 2

Q1: Rank the importance of the following goals when setting CEO pay, by dragging the options below (1=most important, 3=least important). The first row of responses for each option (not italicized) is from directors (n=203), the second row (italicized) is from investors (n=159).

	Mean	1	2	3
Attract/retain the right CEO	1.39 ^{a,a}	65%	32%	4%
	<i>1.64^{a,a}</i>	<i>44%</i>	<i>48%</i>	<i>8%</i>
Design a structure that motivates the CEO	1.70 ^{a,b}	34%	61%	5%
	<i>1.56^{a,b}</i>	<i>51%</i>	<i>42%</i>	<i>7%</i>
Keep the quantum of pay down	2.90 ^{a,b}	1%	8%	91%
	<i>2.80^{a,b}</i>	<i>5%</i>	<i>10%</i>	<i>85%</i>

The first superscript reports whether the mean is significantly different from two (the middle ranking), the second whether the means are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

⁶ To opt in to receive the draft, after completing the survey, participants were invited to add their email address. This final step was optional; approximately half the respondents filled it in. Many respondents were not identifiable from their email.

⁷ The Investment Association, the UK's trade body for the investment industry, classifies 394 funds as UK equity funds. Trustnet provides information on 383 (97%) of these funds.

⁸ Asset manager and asset owners who sign up to the voluntary UK Stewardship Code pledge to exert governance on their investee companies. Nearly all major investors in the UK are signatories, not just ESG investors.

⁹ The results presented are based on all responses, but do not change materially if we only include respondents who answered every question.

¹⁰ “Quantum” is the term most commonly used by UK practitioners for the level of pay, and target quantum is equivalent to expected pay, i.e., salary and benefits plus target bonus and the grant-date value of equity awards.

¹¹ For example, Legal & General Investment Management, the largest UK asset manager, annually publishes “LGIM's UK principles of executive pay.” Every year from 2019 to 2022, the longest section was on quantum and stressed “the current social sensitivities around pay inequality ... the board should consider the wider impact of executive pay, e.g. upon the general workforce, public perception, the economic climate and government bodies.”

65% of directors and 44% of investors view hiring and retention (“attract/retain the right CEO”) as most important, while 34% of directors and 51% of investors rank the structure of pay (“design a structure that motivates the CEO”) most highly. These average rankings are significantly different at the 1% level. This reflects a theme that recurs throughout the survey – directors perceive labor market forces, and thus the IR, as crucial, while investors assign greater importance to the IC.

There are two potential reasons for this divergence. The first is that directors and shareholders have different objectives. One possibility is that shareholders wish to maximize shareholder value but the board is captured by the CEO, or does not wish to exert effort to restructure pay or recruit a new CEO – and cites competitive pressures as an excuse. In addition, because the board is less diversified than institutional investors¹², it may not push for a contract that improves shareholder value but risks leading to a CEO departure. We use “weak boards” to capture all these reasons. A second possibility is that shareholders do not wish to maximize shareholder value but fund flows, which requires reflecting client concerns about CEO pay even if non-value-maximizing.¹³ Alternatively, investors may maximize value across their portfolio and thus take externalities into account – for example, lowering pay at one firm may make it easier to lower pay at others (Acharya and Volpin, 2010; Dicks, 2012).

The second explanation is that both boards and investors have the same objective – shareholder value – but disagree on how to maximize it. This disagreement may stem from two sources. One is “uninformed investors” – investors may not fully understand labor market pressures, the complexity of the CEO job, the value created by a CEO, what motivates a CEO, or the difficulty in replacing a CEO. Boards may better comprehend these issues, potentially because most directors have executive experience, while many investors do not. Several free text responses from directors supported this view. One noted: “Good people are leaving the plc [public limited company] world for private equity in droves. Fund management companies should not throw stones.”¹⁴ The other source of disagreement is “uninformed boards” – directors may underestimate their latitude to restructure pay, or the depth of the CEO labor market in case the CEO leaves as a result. Later responses show that investors believe many boards to be weak or uninformed.

3.2. Constraints

Our next set of questions studies whether directors and investors perceive constraints over and above the IR and IC considered by standard theories. Table 3, Panel A asked directors “How large a sacrifice in shareholder value would you make to avoid controversy on CEO pay?”, and investors the analogous “how large a sacrifice in shareholder value would you tolerate firms making to avoid controversy on CEO pay?”¹⁵

¹² Directors typically serve on only a few boards, so their reputation may be tarnished if their actions lead to a CEO departure, and they bear the time and effort costs of finding a new CEO. In contrast, investors are diversified, so their portfolio is less affected by a CEO departure.

¹³ See Dasgupta et al. (2021) for a survey of the literature on flow-concerned asset managers.

¹⁴ Potentially suggestive of uninformed investors, particularly governance specialists, this is one of the few questions to which fund managers and governance specialists responded differently. 64% of fund managers ranked “attract/retain the right CEO” and 36% “design a structure that motivates the CEO” as most important – almost exactly matching directors.

¹⁵ The question emphasized “sacrifice in shareholder value” because avoiding some controversies may be value enhancing (e.g., avoiding a customer boycott). In beta-testing, we verified that respondents interpreted the question as intended, which was also confirmed by some free-text responses (“creating a balance of commercial reality weighed against ethics and stakeholders is bound to have a degree of controversy” and “doing the right thing” is what motivates our action – and sometimes that’s controversial”).

Table 3

Q2a: How large a sacrifice in shareholder value would you make to avoid controversy on CEO pay? (1=none, 2=small sacrifice, 3=moderate sacrifice, 4=large sacrifice)

Investors: How large a sacrifice in shareholder value would you tolerate firms making to avoid controversy on CEO pay?

Q2b: How important is it to avoid controversy with the following parties? (-2=not at all important, 2=very important).

The first row of responses for each option (not italicized) is from directors (n=197/128), the second row (italicized) is from investors (n=153/84).

Panel A					
	Mean	1	2	3	4
How large a sacrifice in shareholder value would you make to avoid controversy on CEO pay?	1.92	33%	44%	20%	3%
<i>How large a sacrifice in shareholder value would you tolerate firms making to avoid controversy on CEO pay?</i>	<i>1.74</i>	<i>44%</i>	<i>40%</i>	<i>13%</i>	<i>3%</i>

Panel B						
	Mean	-2	-1	0	1	2
Investors	1.24 ^{a,a}	0%	2%	9%	50%	38%
<i>Other investors</i>	<i>0.24^{c,a}</i>	<i>6%</i>	<i>23%</i>	<i>27%</i>	<i>30%</i>	<i>14%</i>
Employees	0.69 ^{a,a}	4%	7%	27%	41%	21%
	1.26 ^{a,a}	0%	6%	12%	32%	50%
Proxy advisors	0.45 ^{a,a}	2%	12%	38%	34%	13%
	-0.12 ^{-a}	10%	32%	29%	20%	10%
Customers	0.20 ^{c,a}	9%	25%	22%	24%	20%
	1.14 ^{a,a}	1%	7%	17%	26%	49%
Policymakers	0.00 ^{-a}	5%	30%	32%	23%	9%
	0.92 ^{a,a}	2%	5%	27%	30%	36%
Media	-0.05 ⁻	6%	27%	37%	23%	5%
	0.17 ⁻	8%	21%	27%	31%	12%

The first superscript reports whether the mean is significantly different from zero, the second whether the means are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

(For brevity, for future questions we will include the analogous wording for investors in only the results table, not the main text.)

Notably, 67% of directors would sacrifice shareholder value to avoid controversy, and 56% of investors would tolerate directors doing so.¹⁶ This suggests that boards and investors feel restricted by far more than the IR and IC. To the respondents who would tolerate a sacrifice, Panel B asked about the sources of controversy: “How important is it to avoid controversy with the following parties?” This, and most remaining questions, are scored according to a Likert scale, with -2 representing “not at all important”, 2 representing “very important”, and 0 being neutral.¹⁷ We will often report results in the form “x%/y”, where x is the percentage of respondents who selected 1 or 2, i.e., important or very important (which we will together refer to as “important” for brevity), and y is the average rating. Also to avoid cumbersome prose, we will sometimes say “our results suggest that x” rather than “our results suggest that directors and investors believe that x”; however, it is important to bear in mind throughout that our survey only reports directors’ and investors’ perceptions.

The results show that 88% of directors deem it important to avoid controversy with investors; this response also received by far the highest average rating (1.24). This finding is surprising. With symmetric information and objectives, controversy with investors would be avoided by

¹⁶ Investors may tolerate directors sacrificing shareholder value, as investors themselves may suffer reputational damage from being associated with a company viewed as paying its CEO excessively.

¹⁷ The response labels for each question are shown in the relevant table. The survey numbered the options 1 to 5, but we report them as -2 to 2, such that 0 is neutral.

maximizing, not sacrificing, shareholder value. Directors would simply offer the optimal contract and investors would agree that it is optimal. Instead, paradoxically, directors believe they have to sacrifice shareholder value to satisfy shareholder requirements. One director wrote that “shareholders appoint RemCos and then often seek to micromanage their duties.” This suggests that directors and investors disagree on what the optimal contract is.

Directors rated employees as the second most important source of controversy (63%/0.69), consistent with concerns about internal equity that we will revisit later. The third highest was proxy advisors (48%/0.45), consistent with their strong influence over shareholder votes documented by Malenko and Shen (2016). Like the “investors” result, this high score is interesting, as proxy advisors should be acting in shareholders’ interest. Instead, they may not be maximizing shareholder value, perhaps because they impose one-size-fits-all rules (Iliev and Lowry, 2015; Cabezón, 2022; Jochem et al. 2021). Indeed, 71% of directors believe that proxy advisors influence pay more than they should (see Table OA5).

Shareholders, on the other hand, believe the main sources of controversy to be employees (82%/1.26), customers (75%/1.14) and policymakers (65%/0.92).¹⁸ This is consistent with high CEO pay causing outrage and with investors viewing such outrage as costly (Jensen and Murphy, 1990; Bebchuk and Fried, 2004).¹⁹ Notably, 44% of shareholders rated “other investors” as an important source of controversy, suggesting that investors disagree with each other on the optimal contract. Both directors and investors ranked the media as least important. Consistent with this, neither Core et al. (2008) nor Kuhnén and Niessen (2012) find an effect of negative press coverage on the level of pay, although the latter show that it affects its composition.

3.3. Constraints and the level of pay

We next explore whether the constraints from avoiding controversy have consequences. Because these questions are about the actual CEO pay contracts offered, we asked them to directors only.

Table 4, Panel A studies whether constraints affect the level of pay (“Have any of the following ever caused you to offer a lower quantum of CEO pay than you would like?”). 77% of directors reported that at least one of these non-standard constraints has forced them to offer lower pay than they would like. 60% offered less pay to avoid the “risk of investor opposition,” and 53% to avoid the “risk of ‘vote against’ recommendation from a proxy advisor.” A minority feel constrained by the existing shareholder-approved pay policy (44%), the need to avoid controversy with employees, the media, customers, or policymakers (37%), or past pay practices (28%).

To the 77% of directors who answered “Yes” to at least one constraint, Panel B asked “Did this lower quantum ever lead to the following consequences?” In many “shareholder value” models, CEOs are at their participation constraint, so the only possible outcome is that the current CEO leaves or (for firms with a vacancy) the firm hires someone cheaper. However, only 7% reported that the CEO left, and 13% that they hired a less expensive CEO.

¹⁸ While prior research has documented the influence of legislation on pay (see the surveys of Murphy (2013) and Edmans et al. (2017)), we are unaware of evidence that the desire to avoid controversy with policymakers affects pay. For example, if UK policymakers believe that a CEO has been overpaid, a House of Commons Select Committee might call up the board chair, RemCo chair and major investors to explain why they allowed this to occur. This happened to UK banks following the financial crisis, and more recently to Persimmon and Carillion following pay scandals.

¹⁹ Bebchuk and Grinstein (2005) argue that a relaxation of outrage constraints due to a booming stock market helps explain the rise in US CEO pay between 1993–2003, and that paying in options avoided outrage as they could be justified as incentive mechanisms.

Table 4

Q12a: Have any of the following ever caused you to offer a lower quantum of CEO pay than you would like? (Y/N) (n=172)

Q12b: Did this lower quantum ever lead to the following consequences? (Y/N) (n=132)

Panel A		
	Yes	No
Risk of investor opposition	60%	40%
Risk of “vote against” recommendation from a proxy advisor	53%	47%
Restrictions from our existing approved pay policy	44%	56%
Risk of controversy with employees, the media, customers, or policymakers	37%	63%
Unwillingness to deviate substantially from how we have paid in the past	28%	72%
Panel B		
	Yes	No
The CEO was less motivated	42%	58%
There were no adverse consequences	41%	59%
We hired a less expensive CEO	13%	87%
The CEO left	7%	93%

By far the most popular responses are not predicted by standard theories. 42% of directors reported that “the CEO was less motivated.” In standard models, motivation depends only on the sensitivity and not the level of pay. When the CEO decides whether to increase effort, she compares the marginal increase in consumption utility ($u'(c)$, where c is consumption) with the marginal cost of doing so. Instead, the free text responses and interviews suggest that pay may matter *per se*, not just for the extra consumption utility, because the CEO compares it to what she believes is fair; if paid unfairly, she is demotivated and reduces effort. As one director said about the consequences of a lower quantum of pay, “[the CEO] still did the job, but his morale was affected negatively.” Another wrote: “There is first a test of pay fairness by the CEO, then after that, it is about building reputation” – reputational concerns incentivize CEOs to perform, but only if they view their pay as fair.

These responses are consistent with Akerlof’s (1982) “gift exchange” efficiency wage model, Akerlof and Yellen’s (1990) “fair wage-effort hypothesis,” and Hertzberg’s (1959) view that fair pay is a “hygiene factor” that demotivates if not provided. They also align with a large literature in labor economics showing that, when employees perceive their wage as unfair, they become demotivated.²⁰

Formally, “fair” pay is pay that exceeds one or more reference points. The CEO’s utility may be of the form $f(w, X)$, where f is a function, w is the CEO’s wage, and X is a vector of reference points that represent what she views as fair. For example, $f(w, X) = w - \sum_{i=1}^n \gamma_i \max(x_i - w, 0)$ would imply that the CEO’s pay must exceed all n reference points, else she perceives it as unfair and suffers a utility loss. The dependence on w highlights how the CEO’s utility depends on the wage *per se*, rather than the consumption it enables. Future questions will shed light on what these reference points are.

The second-most reported effect of lower pay was “there were no adverse consequences” (41%). This high frequency is meaningful – response bias would work against directors admitting that external pressures led to a value-creating action that they would not have taken otherwise. Thus, in many cases, directors had greater latitude to cut pay than they thought, consistent with weak or uninformed boards. There are two caveats to this interpretation. One is that some reductions in CEO pay were caused by the COVID-19 pandemic or by coordinated pressure on executive pensions (see Section 2.3), which reduced pay across the market. Lowering pay in isolation may have adverse consequences. A second is that uncertainty may have made boards’ unwillingness to cut pay ex-ante optimal due to a high (100%–41% = 59%)

²⁰ See Fehr et al. (2009) for a review of this literature.

likelihood of negative effects. This is particularly the case if retaining and motivating the right CEO are more important than the level of pay, as reported in Table 2.

3.4. Constraints and the structure of pay

We next investigate how these non-standard constraints affect the structure of pay. Table 5, Panel A asked “Have any of the following ever caused you to offer an inferior structure of CEO pay to what you would like?” Similar to pay levels, boards perceive binding constraints. 72% of directors were forced to offer an inferior pay structure, mostly because of proxy advisors (54%) or investors (54%).

To the 72% of respondents who selected “Yes” to at least one constraint, Panel B asked “Was the structure inferior in the following ways?” 69% responded that they had to “follow market practice more,” likely to comply with one-size-fits-all rules applied by investors or proxy advisors. For example, one director said his board “could have used more creative non-standard vehicles but didn’t.” This echoes [Cabezon \(2022\)](#), who shows that US pay packages have become more homogenized.

65% of directors were forced to offer less upside for good performance. This is interesting since investors, the main source of constraints, view motivating the CEO as important (Table 2). Investors might be concerned that too much upside has perverse consequences, or that employees or customers will object to high pay even if justified by performance. 57% of directors were forced to use more performance conditions. An interviewee complained that different investors have different objectives, and each asks for a performance condition to reflect theirs.

In free-text entries and interviews, several directors and investors explained that they would prefer to pay CEOs like owners, providing large equity stakes, small annual bonuses, and no long-term incentive plans (LTIPs). This would entail unlimited upside and no performance conditions, violating some of the above constraints. Many investors stated that other investors or proxy advisors would object because such a scheme does not fit their models. One director wrote that “we have held off changing from LTIPs to share award schemes for some of the above reasons.” Another pointed out that private firms use large equity stakes, but when they go public, they are instantly benchmarked to other public firms and have to offer bonuses and LTIPs “because this is what everyone else does.”

Table 5

Q13a: Have any of the following ever caused you to offer an inferior structure of CEO pay to what you would like? (Y/N) (n=170)

Q13b: Was the structure inferior in the following ways? (Y/N) (n=123)

Panel A		
	Yes	No
Risk of “vote against” recommendation from a proxy advisor	54%	46%
Risk of investor opposition	54%	46%
Restrictions from our approved pay policy	40%	60%
Restrictions from regulation or governance codes	36%	64%
Risk of controversy with employees, the media, customers, or policymakers	29%	71%
Unwillingness to deviate substantially from how we have paid in the past	16%	84%
Adverse tax, accounting, or disclosure implications	10%	90%
Panel B		
	Yes	No
We followed market practice more	69%	31%
We offered less upside for good performance	65%	35%
We used (more) performance conditions	57%	43%
We made incentives more long-term	40%	60%
We made incentives more short-term	13%	87%

3.5. Summary

Our questions on the objectives and constraints of pay yield the following conclusions:

1. Directors view attracting and retaining the CEO (satisfying the IR) as the most important goal of pay, while investors believe that motivating the CEO (satisfying the IC) is more important. Both view reducing the level of pay as least important.
2. Boards feel constrained by far more than just the IR and IC. 67% of directors would sacrifice shareholder value to avoid controversy on CEO pay, and 56% of investors would tolerate directors doing so.
3. Directors view avoiding controversy with investors as the main constraint, suggesting that they disagree on the optimal contract. Directors also view proxy advisors as a significant constraint; directors and especially investors view avoiding controversy with employees as important.
4. These additional constraints matter. 77% of directors were forced to offer a lower level of pay. 41% of directors admitted that doing so had no adverse consequences, while 42% stated that it reduced the CEO’s motivation. The latter is inconsistent with standard models, but is consistent with CEOs being demotivated by “unfair” pay.
5. The additional constraints forced 72% of directors to offer an inferior pay structure. This typically involved less tailoring and reduced upside for good performance.

4. The level of pay

This section studies the determinants of the level of pay. We investigate the pay of a new CEO upon appointment (Section 4.1) separately from subsequent increases (Section 4.2) and decreases (Section 4.3). Section 4.4 analyzes whether boards and investors believe that pay can be cut.

4.1. Pay of a new CEO

Table 6 asked “How important are the following factors in determining the target quantum of pay for a new CEO?”, and aims to capture the determinants of the ex-ante expected value of pay.²¹ “The new CEO’s ability” ranked first for both directors (85%/1.29) and investors (90%/1.49), consistent with talent-based models such as [Gabaix and Landier \(2008\)](#) and [Terviö \(2008\)](#). Perhaps less expected were the high responses to “how attractive our firm is to run (e.g. prestige, risk, complexity).”²² Many critics of high pay believe that the CEO job involves little disutility, so it should not be a significant driver of pay. In contrast, this response ranked second for investors (61%/0.61) and third for directors (68%/0.76). In several free-text fields and interviews, directors stressed how difficult the CEO job is – involving long hours, being constantly in the media, and pressures not faced by other executives.

²¹ In the academic literature, the level of pay refers to the ex-ante expected value $E[w]$, which is not easy to translate into practitioner terminology. In beta-testing, the “level” of pay was interpreted as realized pay, which is mechanically linked to factors such as CEO ability, as a high-ability CEO will hit bonus thresholds. “Expected pay” was interpreted as the amount of pay that the CEO believes she deserves. The beta-testing revealed that “target quantum” best captured $E[w]$. At the start of the “Level of Pay” section, we stated “In this survey, the quantum of pay should be taken to refer to the target level of total remuneration set by the remuneration committee.”

²² Prestige and complexity aim to capture both the non-pay benefits and difficulties, i.e., the net disutility, of the CEO’s job. Both free-text fields and interviews only referred to prestige and complexity, not risk. This is consistent with [Focke et al. \(2017\)](#) and [Albuquerque et al. \(2023\)](#), who find, respectively, that CEOs of prestigious firms earn less and that risk has little effect on the level of pay.

Table 6

Q3: How important are the following factors in determining the target quantum of pay for a new CEO? (-2=not at all important, 2=very important)
The first row of responses for each option (not italicized) is from directors (n=188), the second row (italicized) is from investors (n=145).

	Mean	-2	-1	0	1	2
The new CEO's ability	1.29 ^{a,b}	4%	2%	9%	31%	54%
	<i>1.49^{a,b}</i>	<i>0%</i>	<i>3%</i>	<i>7%</i>	<i>27%</i>	<i>63%</i>
CEO pay at peer firms	0.82 ^{a,a}	1%	8%	24%	41%	26%
	<i>0.46^{a,a}</i>	<i>5%</i>	<i>9%</i>	<i>37%</i>	<i>33%</i>	<i>16%</i>
How attractive our firm is to run (e.g. prestige, risk, complexity)	0.76 ^{a,-}	4%	7%	22%	45%	22%
<i>How attractive the firm is to run (e.g. prestige, risk, complexity)</i>	<i>0.61^{a,-}</i>	<i>3%</i>	<i>10%</i>	<i>26%</i>	<i>43%</i>	<i>17%</i>
The new CEO's other employment options	0.55 ^{a,b}	6%	11%	26%	38%	20%
	<i>0.26^{a,b}</i>	<i>6%</i>	<i>16%</i>	<i>34%</i>	<i>32%</i>	<i>11%</i>
The new CEO's pay in their previous position	0.26 ^{a,a}	5%	15%	38%	32%	9%
	<i>-0.21^{a,a}</i>	<i>10%</i>	<i>26%</i>	<i>41%</i>	<i>20%</i>	<i>3%</i>
How financially motivated the new CEO is	0.07 ^{-b}	9%	20%	35%	30%	7%
	<i>-0.23^{b,b}</i>	<i>14%</i>	<i>23%</i>	<i>39%</i>	<i>19%</i>	<i>5%</i>
The outgoing CEO's pay	-0.02 ^{-a}	7%	26%	35%	28%	5%
	<i>-0.55^{a,a}</i>	<i>18%</i>	<i>37%</i>	<i>30%</i>	<i>11%</i>	<i>4%</i>

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

“CEO pay at peer firms” was the second most popular response for directors (66%/0.82) and third for investors (49%/0.46). Notably, both ranked it higher than “the new CEO's pay in their previous position” (42%/0.26 for directors and 23%/-0.21 for investors) and “the new CEO's other employment options” (58%/0.55 and 43%/0.26), which more directly reflect the CEO's outside alternatives. This suggests that pay at peer firms is important beyond the participation constraint, and is relevant even if the CEO could not move to those firms. Several interviewees argued that CEOs use peer comparison to deduce the *worth* the board and investors attach to them. One director wrote that “relative competition (why does he earn more than me?) is very significant as an issue of pride.” Thus, peer pay may determine what the CEO views as fair and be a reference point included in *X*. A potential test of this channel is whether CEO pay responds to changes in pay at peer firms that the CEO is unlikely to move to, e.g., due to there being no vacancies. Investors viewed all three of the above determinants as less important than directors, consistent with them placing less weight on labor market conditions.

The free-text fields highlighted an additional determinant of pay absent from most models – internal equity considerations. One director

emphasized the importance of “the multiple of the CEO pay to the average within the company”; another responded that “internal proportionality is most important.” Several investors made similar statements.

4.2. Increases in pay for an incumbent CEO

Table 7 asked “What causes you to increase the target quantum of pay for an incumbent CEO?” As with the prior question, there was general agreement between directors and investors. Both rated “good recent CEO performance” the highest, with support of 76%/0.98 from directors and 75%/1.05 from investors. While intuitive, it may seem unnecessary to reward high performance with increased pay, given that almost all CEOs have substantial equity holdings. Empirically, the incentives from changes in pay levels are so small compared to those from CEOs' equity holdings (Hall and Liebman, 1998) that they are typically ignored when calculating incentives.

The free text responses and interviews suggest that pay rises are used to provide *ex-post recognition* of performance. A fund manager explained in an interview that pay rises are important to acknowledge good

Table 7

Q4: What causes you to increase the target quantum of pay for an incumbent CEO? (-2=strongly disagree, 2=strongly agree)

Investors: What causes you to support increases to the target quantum of pay for an incumbent CEO?

The first row of responses for each option (not italicized) is from directors (n=186), the second row (italicized) is from investors (n=142).

	Mean	-2	-1	0	1	2
Good recent CEO performance	0.98 ^{a,-}	3%	7%	15%	41%	34%
	<i>1.05^{a,-}</i>	<i>0%</i>	<i>5%</i>	<i>20%</i>	<i>41%</i>	<i>35%</i>
Increase in firm size	0.37 ^{a,c}	4%	15%	35%	33%	13%
	<i>0.17^{c,-}</i>	<i>9%</i>	<i>15%</i>	<i>30%</i>	<i>39%</i>	<i>6%</i>
Increase in pay at peer firms	0.26 ^{a,a}	3%	16%	37%	38%	5%
	<i>-0.17^{b,a}</i>	<i>13%</i>	<i>20%</i>	<i>40%</i>	<i>25%</i>	<i>2%</i>
Increased threat of CEO leaving	0.25 ^{a,a}	5%	17%	35%	35%	8%
	<i>-0.06^{-a}</i>	<i>8%</i>	<i>24%</i>	<i>38%</i>	<i>27%</i>	<i>3%</i>
Changes in attractiveness (e.g. prestige, risk, complexity) of CEO job at your firm	0.23 ^{a,-}	6%	17%	33%	34%	9%
<i>Changes in attractiveness (e.g. prestige, risk, complexity) of CEO job at their firm</i>	<i>0.25^{a,-}</i>	<i>8%</i>	<i>13%</i>	<i>34%</i>	<i>36%</i>	<i>9%</i>
Other changes that reduce the attractiveness of the pay package (e.g. adding holding periods)	-0.11 ^{-,-}	8%	26%	38%	25%	3%
	<i>0.01^{-,-}</i>	<i>7%</i>	<i>23%</i>	<i>40%</i>	<i>21%</i>	<i>8%</i>
Changes in attractiveness (e.g. prestige, risk, complexity) of CEO jobs at other firms	-0.28 ^{a,-}	10%	30%	41%	16%	3%
	<i>-0.37^{a,-}</i>	<i>10%</i>	<i>34%</i>	<i>40%</i>	<i>15%</i>	<i>1%</i>

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

performance, and that no talented person stays in a job where she does not feel appreciated. Another pointed out that a pay rise is so small compared to firm value that it costs the company very little. Thus, if the CEO is denied it, she infers that the board and shareholders do not value her highly and may be demotivated or leave.²³

These findings help us refine the notion of fairness used by CEOs. Existing research on fairness does not differentiate between flow pay and changes in wealth: fairness models are silent about where a fair reward should come from; in fairness experiments, subjects have no pre-existing wealth and only receive flow pay. If flow pay and changes in wealth are fungible, fairness concerns would not necessitate pay increases, since a CEO's equity already rises upon good performance.

Our results suggest that changes in flow pay play a special role by addressing CEOs' desire for *recognition*. A pay increase, unlike the revaluation of equity, provides internal recognition because it requires a discretionary decision by the board and approval by shareholders. It provides external recognition because it is observable and thus boosts the CEO's reputation. Several interviewees explained that a pay rise is a more public endorsement than the revaluation of equity, since it is disclosed as part of the "single figure."²⁴ An empirical implication is that discretionary pay awards should follow good performance even if their effects on consumption utility are negligible. For example, Larry Ellison owns over \$100 billion of Oracle stock and earns over \$1 billion per year in Oracle dividends, yet receives a multi-million dollar cash bonus in a typical year. In addition, the new US requirement to disclose "Compensation Actually Paid", which includes changes in the fair value of unvested awards, may increase the recognition provided by previously-granted equity and reduce the role of flow pay.

These responses also suggest that the CEO's perception of her contribution to the firm determines another reference point included in X . This is also a potential micro-foundation for why CEOs might have higher discount rates than shareholders – not because they need current pay for consumption, but because changes in pay affect their perceptions of fairness and recognition.

All other responses received less than 50% support from both groups. The second most common response from directors was "increases in firm size" (46%/0.37; 45%/0.17 from investors). Some free-text comments and interviewees argued that size matters because of complexity, consistent with the findings of Gayle and Miller (2009). Others explained that CEOs benchmark their pay against peers of similar size to assess whether it is fair. As one director argued, "benchmarking gives comfort that there's fairness, transparency, and objectivity." This might be because firm size and peer pay are more observable than other potential determinants of pay, such as CEO talent or disutility, so linking pay to them gives the CEO comfort that pay is set fairly.

Directors ranked "increases in pay at peer firms" (44%/0.26) significantly higher than investors (27%/0.17). Because CEOs are rarely hired away (Cziraki and Jenter, 2022), this is more likely due to directors being attuned to the importance of fairness in motivating CEOs rather than labor market pressures. Some investors were strongly opposed to responding to increases in peer firm pay. Both groups considered peer pay far less important for changes in an incumbent's pay than for the pay of a new CEO. This is consistent with Kahneman et al.'s (1986) evidence on employees' reference point for pay fairness – peer firm pay for new hires, and last year's pay for incumbents.

²³ Another explanation is to retain the CEO if good performance increases her outside options. However, the support for "increased threat of CEO leaving" (43%/0.25 for directors and 30%/0.06 for investors) was much lower.

²⁴ Firms are required to disclose total realized pay, known as the "single figure," which includes salary, annual bonus, and cash and equity earned as part of LTIPs. Equity earned is based on when performance conditions are tested and may not yet be paid out, since there may be additional holding periods. New grants of equity are disclosed separately, together with the performance conditions that govern their vesting.

The low responses to "other changes that reduce the attractiveness of the pay package (e.g. adding holding periods)" (25%/0.11 for directors and 30%/0.01 for investors) are surprising since, in theory, it is the value of the entire package that needs to exceed the CEO's outside option. An interviewee explained that the components of CEO pay are negotiated in isolation, consistent with narrow framing. The board and CEO negotiate base salary first, then the incentive components, and then restrictions such as holding periods. It is therefore unlikely for the CEO to accommodate a board's demand for longer holding periods by asking for an increase in the already-negotiated base salary.

Several free-text entries stressed that significant increases in the level of CEO pay are difficult. Many directors believe that they risk controversy, unless justified by a visible change to the size or complexity of the firm. One explained that "It is pretty much impossible to increase the target pay of an incumbent CEO in the UK. It might be possible with a large acquisition that changes the scale and complexity substantially"; another stated "substantial increases would only come if the job gets markedly bigger/more complex."

In the same vein, and consistent with the free-text responses accompanying Table 6, several free-text entries argued that it is difficult to increase CEO pay much faster than overall workforce pay. This suggests that fairness also matters for directors and investors, either because they themselves believe that a disproportionate increase would be unfair, or because employees or external stakeholders would view it as unfair and respondents want to avoid controversy. Thus, their perceived cost of compensation may be $w + g(w, Y)$ where g is a function and Y is a vector of reference points (different from X), such as worker pay, CEO pay in peer firms, or last year's pay. Consistent with this, Pan et al. (2022) show that equity markets reacted negatively to the first-time disclosure of high pay ratios by US firms.

4.3. Decreases in pay for an incumbent CEO

We now turn to decreases in pay. We first asked directors in Table 8, Panel A, "Have you ever significantly decreased the target quantum of pay for an incumbent CEO?" Only 23% of directors responded "Yes." Combined with the difficulty of significantly increasing pay, this suggests that CEO pay is sticky, and that directors decide pay changes rather than pay levels, analogous to the findings of Lintner (1956) and Brav et al. (2005) for dividends. We interviewed directors on why pay cuts are so rare. One stated that if there were ever a justification for cutting CEO pay, you would instead fire her. Another explained that a pay cut communicates that the board has lowered its assessment of the CEO, and a third said that, as a result, cutting a CEO's pay is effectively firing her. This suggests that another reference point in X is prior year pay, consistent with Kahneman et al.'s (1986) evidence.

To the respondents who answered "Yes," Panel B asked "What caused you to decrease the target quantum of pay for an incumbent CEO?" Here, unlike for increases, there was significant disagreement between investors and directors. The five factors that standard theories predict should drive reductions in pay (poor performance, decreases in firm size, changes in attractiveness at your firm, changes in attractiveness at other firms, decreases in pay at peer firms) were the five least popular options for directors, all with negative average ratings. In contrast, investors believe that these five factors should drive both increases and decreases in pay, as theory predicts.

One potential explanation for this disagreement is that investors are less concerned about the motivational or retention consequences of pay cuts. Investors' most popular reason to cut pay was poor CEO performance (70%/0.96), even though such a cut may erode the CEO's sense of worth. CEOs may also view pay cuts because of changes at peer firms or their own job getting easier as unfair, as these changes are out of their control. Recall, however, that directors view increases in peer firm pay and job complexity as valid justifications for pay rises. This asymmetry echoes Garvey and Milbourn (2006), who find that CEOs are rewarded for good luck but not penalized for bad luck.

Table 8

Q5a: Have you ever significantly decreased the target quantum of pay for an incumbent CEO? (Y/N)

Investors: Have you ever requested significant decreases to the target quantum of pay for an incumbent CEO?

Q5b: What caused you to decrease the target quantum of pay for an incumbent CEO? (-2=strongly disagree, 2=strongly agree)

Investors: What caused you to request decreases to the target quantum of pay for an incumbent CEO?

The first row of responses for each option (not italicized) is from directors (n=186/41), the second row (italicized) is from investors (n=142/93).

Panel A						
		Yes		No		
Have you ever significantly decreased the target quantum of pay for an incumbent CEO?		23%		77%		
<i>Have you ever requested significant decreases to the target quantum of pay for an incumbent CEO?</i>		65%		35%		
Panel B						
	Mean	-2	-1	0	1	2
External pressure to reduce pay (e.g. from investors, the media, policymakers)	0.15 ^{-,c}	20%	10%	24%	29%	17%
	-0.28 ^{ab,c}	19%	20%	33%	23%	4%
Your firm encountering financial constraints	0.07 ^{-,}	32%	12%	5%	20%	32%
<i>The firm encountering financial constraints</i>	0.53 ^{a,-}	13%	9%	20%	29%	29%
The CEO requesting it*	-0.37 ^{-,}	49%	0%	10%	22%	20%
Poor recent CEO performance	-0.41 ^{-,a}	41%	10%	15%	17%	17%
	0.96 ^{a,a}	6%	6%	17%	25%	45%
Decrease in firm size	-0.90 ^{ab,b}	46%	20%	17%	12%	5%
	-0.31 ^{b,b}	23%	18%	35%	15%	9%
Decrease in pay at peer firms	-1.15 ^{a,c}	56%	17%	15%	10%	2%
	-0.74 ^{a,c}	29%	28%	32%	10%	1%
Change in attractiveness (e.g. prestige, risk, complexity) of CEO job at your firm	-1.20 ^{a,a}	56%	17%	17%	10%	0%
<i>Change in attractiveness (e.g. prestige, risk, complexity) of CEO job at their firm</i>	-0.51 ^{a,a}	22%	28%	34%	12%	4%
Change in attractiveness (e.g. prestige, risk, complexity) of CEO jobs at other firms	-1.41 ^{a,a}	63%	20%	12%	5%	0%
	-0.80 ^{a,a}	28%	35%	28%	5%	3%

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively. *Only directors were offered this response.

Table 9

Q6: If your firm reduced the target quantum of pay of its next CEO by 1/3 compared to its current CEO, what might happen? (-2=very unlikely outcome, 2=very likely outcome)

Investors: If a firm reduced the target quantum of pay of its next CEO by 1/3 compared to its current CEO, what might happen?

The first row of responses for each option (not italicized) is from directors (n=182), the second row (italicized) is from investors (n=140).

	Mean	-2	-1	0	1	2
We would recruit a lower quality CEO	0.66 ^{a,a}	7%	9%	25%	30%	29%
<i>The firm would recruit a lower quality CEO</i>	<i>-0.32^{a,a}</i>	<i>11%</i>	<i>30%</i>	<i>41%</i>	<i>15%</i>	<i>3%</i>
The CEO would be less motivated	0.38 ^{a,a}	4%	19%	31%	26%	20%
<i>The CEO would be less motivated</i>	<i>-0.31^{a,a}</i>	<i>15%</i>	<i>27%</i>	<i>34%</i>	<i>21%</i>	<i>2%</i>
It would create undesirable pay compression between the CEO and other executives	0.36 ^{a,a}	7%	16%	26%	37%	14%
<i>It would create undesirable pay compression between the CEO and other executives</i>	<i>-0.50^{a,a}</i>	<i>16%</i>	<i>36%</i>	<i>31%</i>	<i>14%</i>	<i>2%</i>
We would have a strained relationship with the CEO	0.32 ^{a,a}	7%	19%	29%	26%	19%
<i>The board would have a strained relationship with the CEO</i>	<i>-0.48^{a,a}</i>	<i>14%</i>	<i>34%</i>	<i>39%</i>	<i>9%</i>	<i>3%</i>
It would send a negative signal about CEO quality to the market	0.29 ^{a,a}	8%	19%	24%	36%	14%
<i>It would send a negative signal about CEO quality to the market</i>	<i>-0.36^{a,a}</i>	<i>14%</i>	<i>33%</i>	<i>32%</i>	<i>16%</i>	<i>5%</i>
There would be no adverse consequences	-0.96 ^{a,a}	41%	26%	23%	7%	3%
<i>There would be no adverse consequences</i>	<i>-0.02^{-a}</i>	<i>9%</i>	<i>23%</i>	<i>35%</i>	<i>26%</i>	<i>6%</i>

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

Directors' three most common reasons to cut pay had no analogy in the question about increases. The most popular response (46%/0.15, versus 27%/-0.28 for investors) was "external pressure to reduce pay." The free-text responses highlighted two recent causes of external pressure – the COVID-19 pandemic, and the controversy about executive pensions. A second popular reason was "your firm encountering financial constraints" (51%/0.07 for directors, 58%/0.53 for investors).²⁵

²⁵ This response differed according to firm size (35%/-0.50 for directors in the FTSE 100 versus 78%/1.00 for the FTSE Small Cap), consistent with tighter financial constraints and CEO pay being a larger fraction of profits in smaller firms.

A third, and unexpected, justification to cut pay was "the CEO requesting it" (41%/-0.37).²⁶ While contrary to almost any model, it can be justified by the CEO having a sense of fairness or wanting to pre-empt controversy. One director wrote: "I was the incumbent CEO and asked for a reduction to reflect reduced complexity and intensity of role." Another director said that it should not be up to the board to be tough on pay; a CEO should be tough on her own pay to show that she is sensitive to the environment. Taking these three reasons together, directors see pay cuts as plausible if they can be attributed to external pressures or

²⁶ We did not offer this response to investors as the CEO would make such a request to the board, not to them.

financial constraints, or are requested by the CEO. Unlike cuts in response to bad performance, such cuts do not challenge the CEOs' sense of her worth.

In their free-text responses, several investors stressed another justification for pay decreases absent from standard models – that pay is simply too high, and thus cuts are justified even without any change in circumstances: “the total compensation is unreasonably high”, “the quantum of pay was insanely high relative to anything – a formula gone wrong.” In shareholder value models, pay is always set optimally, so it should only change when parameters change. In contrast, many investors believe that pay was not optimal to begin with. There were no such responses from directors.

4.4. Why is CEO pay so high?

Our final questions on the level of pay aim to understand why CEO pay is so high. This is difficult to ask directly, as directors may claim CEO ability or market forces as justification. We thus asked “If your firm reduced the target quantum of pay of its next CEO by 1/3 compared to its current CEO, what might happen?” We focused on CEO changes as this is when a board has greatest flexibility to reset pay. We chose a cut of 1/3 because it is larger than typical reductions in a transition yet not entirely implausible. In the vast majority of companies, it would still leave the CEO the highest-paid employee.

The responses, reported in Table 9, show a significant disparity between directors and investors. Only 10% of directors agreed that “there would be no adverse consequences” (average score -0.96). Their strongest concern was “we would recruit a lower quality CEO” (59%/0.66). Several free text entries emphasized that reducing the pay of one firm in isolation without corresponding reductions in the market would be dangerous. This indicates that directors believe the CEO labor market to be competitive and the talent supply to be limited, consistent with prior evidence that many firms struggle to replace CEOs who unexpectedly depart (e.g., Jenter et al. 2022).

Directors' second greatest concern was “the CEO would be less motivated” (46%/0.38). This is consistent with Table 4, Panel B, and again suggests that the level of pay has motivational consequences missing from standard models. Just over half agreed that “it would create undesirable pay compression between the CEO and other executives” (51%/0.36), another concern absent from most models, which focus on CEO pay in isolation.

One potential explanation is that compression reduces tournament incentives (Lazear and Rosen, 1981; Kale et al. 2009). However, free-text fields and interviews cited this reason only once. Instead, they

suggested that pay disparity is required for fairness. One interviewee explained that “the more you do for the company, the more you should be paid; if the CEO isn't doing more, she shouldn't be CEO.” Another highlighted that CEOs are under more pressure and public scrutiny than other executives. A third explained that reduced pay disparity would “disrupt the natural order of things and the hierarchies within the organization,” and that this applies beyond the CEO. If a CFO were paid less than the commercial director, even if dictated by market forces, the finance function would view itself as less important. Thus, while prior responses suggest that directors want CEO pay to not exceed certain levels, such as a multiple of worker pay, these additional fairness concerns imply that pay should also not fall short of other reference points, such as a certain premium to other top executives.

In contrast, investors are much more open to cutting CEO pay than directors. Their most popular response was that “there would be no adverse consequences” (33%/-0.02, with 35% neither agreeing nor disagreeing). Fewer than a quarter agreed with each of the negative consequences. Thus, many investors believe it is possible to reduce CEO pay by as much as 1/3. Several explained that, even if the board ends up recruiting a different CEO, she would be less materialistic rather than less capable. A fund manager claimed that “CEOs should not just be motivated by quantum of compensation – that suggests they have the wrong person;” a governance specialist stated that “[the CEO] might have a hissy fit ... then the board should reconsider if this person is appropriate for the role.”

Investors' view that CEO pay is often unjustified was also reflected in their responses to “Do you believe the overall level of CEO pay is too low, too high, or about right?”, reported in Table OA4. 77% of investors view pay as too high. We asked these 77% why it is too high, and 86% agreed that “boards are ineffective at lowering it even though they should.” Thus, many investors view CEO pay as excessive, and they blame boards for it.

4.5. Summary

Our questions on the level of pay yield the following conclusions:

1. Directors believe that a significantly lower level of pay would markedly worsen the quality and motivation of the CEO. Lower pay reduces motivation because the CEO perceives it as unfair, in contrast to standard models.
2. Many investors, in contrast, see few adverse consequences of lowering pay, even if done by one firm in isolation. Instead, they view a CEO transition as an opportunity to reset the level of pay. 77%

Table 10

Q7: What motivates your CEO to perform strongly? (-2=not at all important, 2=very important)

Investors: What motivates CEOs to perform strongly?

The first row of responses for each option (not italicized) is from directors (n=181), the second row (italicized) is from investors (n=139).

	Mean	-2	-1	0	1	2
Intrinsic motivation	1.55 ^{a,-}	2%	1%	4%	24%	68%
	<i>1.50^{a,-}</i>	<i>0%</i>	<i>1%</i>	<i>9%</i>	<i>31%</i>	<i>60%</i>
Personal reputation	1.40 ^{a,b}	3%	1%	6%	36%	55%
	<i>1.60^{a,b}</i>	<i>0%</i>	<i>0%</i>	<i>4%</i>	<i>33%</i>	<i>63%</i>
Incentives from bonuses, LTIPs, equity, or future pay increases	0.98 ^{a,-}	2%	6%	17%	46%	30%
	<i>0.83^{a,-}</i>	<i>1%</i>	<i>8%</i>	<i>24%</i>	<i>42%</i>	<i>25%</i>
Industry competition	0.62 ^{a,c}	2%	13%	24%	44%	18%
	<i>0.82^{a,c}</i>	<i>1%</i>	<i>6%</i>	<i>26%</i>	<i>44%</i>	<i>23%</i>
The quantum of pay	0.55 ^{a,a}	2%	12%	31%	40%	15%
	<i>0.21^{a,a}</i>	<i>3%</i>	<i>17%</i>	<i>42%</i>	<i>31%</i>	<i>6%</i>
The potential to move to a bigger firm	-0.53 ^{a,a}	19%	35%	29%	16%	2%
	<i>0.37^{a,a}</i>	<i>4%</i>	<i>13%</i>	<i>37%</i>	<i>35%</i>	<i>12%</i>
Risk of being fired	-0.88 ^{a,a}	30%	41%	18%	9%	2%
	<i>-0.20^{b,a}</i>	<i>11%</i>	<i>27%</i>	<i>37%</i>	<i>22%</i>	<i>3%</i>

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

Table 11

Q8: Why do you offer the CEO variable pay? (-2=strongly disagree, 2=strongly agree)

Investors: Why should CEOs be offered variable pay?

The first row of responses for each option (not italicized) is from directors (n=179), the second row (italicized) is from investors (n=137).

	Mean	-2	-1	0	1	2
To motivate the CEO to improve long-term shareholder value	1.46 ^{a,-}	2%	2%	7%	26%	63%
	1.36 ^{a,-}	1%	4%	8%	31%	55%
To attract/retain a high-ability or hard-working CEO	1.19 ^{a,a}	2%	2%	8%	49%	39%
	0.85 ^{a,a}	4%	7%	19%	37%	32%
So that the CEO shares risks with investors and stakeholders, even if out of the CEO's control	1.16 ^{a,-}	3%	2%	10%	44%	41%
	1.14 ^{a,-}	4%	4%	14%	33%	46%
To motivate the CEO to improve outcomes other than long-term shareholder value	0.46 ^{a,-}	9%	13%	26%	27%	25%
	0.47 ^{a,-}	10%	13%	23%	26%	27%
To match peer firm practice	0.37 ^{a,a}	5%	14%	32%	36%	12%
	-0.45 ^{a,a}	15%	31%	39%	12%	3%
Because investors or proxy advisors require it*	-0.17 ^c	18%	20%	31%	23%	8%
So that the quantum of pay can be justified	-0.42 ^{a,-}	22%	30%	21%	21%	6%
	-0.34 ^{a,-}	23%	26%	26%	11%	14%

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively. *Only directors were offered this response.

of investors believe that pay is too high, which they attribute to boards being weak.

- Both investors and directors believe that good recent performance should increase expected pay, even though most CEOs have substantial incentives from their equity holdings. Pay increases are important because CEOs care about the ex-post recognition of their performance.
- Directors view themselves as not starting from a blank sheet of paper. They believe it is difficult to increase pay, and even harder to decrease pay, in response to firm-specific changes. In contrast, investors believe that such changes should drive both increases and decreases in pay.
- Ability is seen as the most important determinant of a new CEO's pay. The (dis)utility of a CEO job is another important driver, suggesting significant cross-sectional variation.
- Pay in peer firms affects new CEO pay even more than the CEO's alternative employment opportunities and the pay at her prior job. This is because peer firm pay affects what the CEO views as fair, rather than because of labor market competition.
- CEO pay levels are affected by internal comparisons. Directors and investors believe there should be a substantial gap to other top executives, but also that the gap to the wider workforce should not be too large.

5. The structure of pay

This section studies the structure of pay. Section 5.1 asks what directors and investors believe motivates a CEO, Section 5.2 why companies offer variable pay, and Section 5.3 how they determine the split between fixed and variable pay. Sections 5.4 and 5.5 explore why two key predictions of theory are not always implemented – long-term incentives and relative performance evaluation.

5.1. What motivates a CEO?

Table 10 asked “What motivates your CEO to perform strongly?” Most models assume that financial incentives are the only motivator. However, while the support for “incentives from bonuses, LTIPs, equity, or future pay increases” was high (76%/0.98 for directors and 68%/0.83 for investors), it was only the third highest-rated response from both groups.

Instead, directors and investors agree that by far the strongest drivers of CEO effort are “intrinsic motivation” (92%/1.55 for directors and 91%/1.50 for investors) and “personal reputation” (91%/1.40 and 96%/

1.60). Both are absent from the vast majority of contract theories.²⁷ Personal reputation scored highly even though we offered separate responses capturing the financial benefits of an improved reputation (pay increases and labor market consequences). This suggests that simply being seen to do a good job is important. An investor said that “There is a common misperception that pay or career progression motivates CEOs to do a good job. [Performance] comes from intrinsic motivation, passion for the job, and maybe a concern with reputation.”

The free-text responses suggest that the main purpose of incentive pay is to provide ex-post recognition of performance, and that incentive pay interacts with intrinsic motivation and reputational concerns (we will use “intrinsic incentives” as an umbrella term for both). The CEO values incentive pay not because she needs it to afford consumption – as one investor stressed, “all CEOs are going to take care of most human needs in terms of finances” – but because it provides recognition that she has performed well. One director stated that “primary motivation comes from inside, but pay is important as a signal to the CEO and the market of the value placed on them by the board.”

“The quantum of pay” (55%/0.55 for directors, 37%/0.21 for investors) is seen as another important motivator, inconsistent with most theory models but consistent with the importance of fairness. One director explained that “principle and sense of fairness tends to matter a lot.”

Finally, even though career concerns have been a focus of the literature on CEO incentives since Fama (1980), they are not seen as important, especially by directors. “The potential to move to a bigger firm” received support of only 18%/-0.53 from directors (46%/0.37 from investors), and the “risk of being fired” of only 11%/-0.88 (25%/-0.20).²⁸ The former is consistent with the rarity of incumbent

²⁷ Some exceptions exist. In Carlin and Gervais (2009), the agent is intrinsically motivated to increase firm value; in Besley and Ghatak (2005), Bettignies and Robinson (2018), and Murdock (2002), effort generates social welfare, which the agent is intrinsically motivated to increase. Bénabou and Tirole (2003) study how extrinsic incentives affect intrinsic motivation, but not the reverse. Under a broad interpretation, a low cost of effort could capture high intrinsic motivation, but it does not capture any intrinsic drive of the CEO to achieve specific goals.

²⁸ The latter result is not due to CEO turnover in the UK being unusually low. Burns et al. (2023) find a CEO turnover rate of 14% for large publicly traded UK firms in 2004–2016, while Jenter and Lewellen (2021) report 12% for US S&P 1,500 firms in 1993–2011. In 2018, the rate in large UK companies was 14% (Robert Half: “CEO turnover in decline as internal candidates rise to the top”) vs 14.7% in North America (PwC: “CEO turnover at record high”).

Table 12

Q9: What determines the split between fixed and variable pay? (-2=not at all important, 2=very important)

Investors: What should determine the split between fixed and variable pay?

The first row of responses for each option (not italicized) is from directors (n=178), the second row (italicized) is from investors (n=135).

	Mean	-2	-1	0	1	2
How much the CEO can affect firm performance	0.66 ^{a,a}	7%	9%	21%	35%	27%
	<i>0.98^{a,a}</i>	<i>1%</i>	<i>5%</i>	<i>19%</i>	<i>44%</i>	<i>31%</i>
Investor or proxy advisor expectations*	0.55 ^a	7%	7%	26%	43%	17%
The split between fixed and variable pay in peer firms	0.37 ^{a,a}	6%	10%	35%	38%	11%
	<i>-0.49^{a,a}</i>	<i>16%</i>	<i>28%</i>	<i>45%</i>	<i>9%</i>	<i>1%</i>
CEO intrinsic motivation	0.30 ^{a,-}	8%	13%	29%	37%	12%
	<i>0.12^{a,-}</i>	<i>13%</i>	<i>16%</i>	<i>32%</i>	<i>25%</i>	<i>14%</i>
The desire to avoid excessive pay outcomes	0.24 ^{a,-}	10%	19%	21%	38%	12%
	<i>0.30^{a,-}</i>	<i>7%</i>	<i>21%</i>	<i>24%</i>	<i>30%</i>	<i>18%</i>
How risky our firm is	-0.44 ^{a,a}	17%	29%	38%	12%	3%
	<i>0.29^{a,a}</i>	<i>7%</i>	<i>13%</i>	<i>32%</i>	<i>38%</i>	<i>10%</i>
CEO personal risk appetite	-0.46 ^{a,-}	20%	31%	27%	19%	3%
	<i>-0.45^{a,-}</i>	<i>19%</i>	<i>30%</i>	<i>32%</i>	<i>19%</i>	<i>1%</i>

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively. *Only directors were offered this response.

CEOs being hired away (Cziraki and Jenter, 2022).²⁹ On the latter, one interviewee explained that boards hire CEOs that are confident in their ability to succeed, so they are not motivated by firing risk.

5.2. The motivation for variable pay

Table 11 asked “Why do you offer the CEO variable pay?” Practitioners interpret “variable pay” as variable flow pay, and thus do not include the CEO’s equity holdings. The most popular response was “to motivate the CEO to improve long-term shareholder value” (89%/1.46 for directors and 87%/1.36 for investors). Thus, even though both believe that intrinsic incentives are the main motivators, financial incentives remain important. Free-text fields and interviews suggested two reasons. First, financial incentives may reinforce intrinsic incentives, in contrast to common arguments that they crowd them out. As one interviewee explained, only a “superhuman” CEO would perform at her very best without financial incentives.³⁰ Second, intrinsic incentives may lead to CEO actions that decrease firm value. Examples given include increasing the scale of the business, R&D for scientific curiosity, or designing the highest-quality product even if a low-cost strategy would be better.³¹

The popularity of this response is interesting also because many CEOs have substantial equity holdings. Thus, even if financial incentives are needed, it is not clear why they need to be provided by variable pay. Interviews and free text fields provided three explanations. First, pay incentives are different from portfolio incentives because they provide recognition. Directors explained that variable pay is used “so the CEO would directly and quickly feel the impact of good and (where relevant) poor short-term performance” and “to recognize achievement – the retrospective acknowledgement of exceptional performance is important.”

²⁹ Surprisingly, upward moves are seen as even less important (7%/-0.62) by FTSE Small Cap directors, even though CEOs of smaller firms are more likely to be hired away.

³⁰ Hartzell et al. (2010) show that even church clergy receive incentive pay, despite intrinsic motivation being presumably strong.

³¹ Our respondents may also support financial incentives because they do not view them as especially expensive. Aside from crowding out or external constraints, financial incentives may be costly for two reasons. The first is limited liability rent, but high CEO salaries suggest that the CEO’s limited liability constraint is not binding. The second is the risk premium required. However, Table 12 will show that CEO risk aversion and firm risk were the least popular responses for the determinants of the split between fixed and variable pay, suggesting the risk premium is minor.

This mechanism suggests that the role of incentive pay may be different to that assumed in standard theories. In such models, variable pay at $t=1$ provides ex ante effort incentives at $t=0$. Our results suggest that variable pay may have an additional role – it ensures that the CEO is recognized at $t=1$ for her actions at $t=0$, so that she remains intrinsically motivated to exert effort at $t=1$.³²

The second reason is that variable pay is given to other employees. One director pointed out that “variable pay is an organization wide practice ... difficult to think of CEO scheme in isolation,” another that “a high proportion of variable pay runs through all levels of the organization (commission at lower levels, bonuses at higher levels) and it therefore feels appropriate and a cultural alignment for the CEO to have a high mix of variable pay.” While internal equity comparisons are typically about the level of pay (e.g., pay ratios), the comparison of pay structures has received less attention.

The third reason is that variable pay can be based on criteria other than the stock price. One director explained how CEOs are set business plans with key performance indicators (“KPIs”), and tying pay to these KPIs holds management accountable. He also noted that CEO actions have a greater effect on these KPIs than on the stock price. Another director argued that the company’s KPIs are only credible to employees and investors if they are incorporated into the CEO’s contract, and that doing so is important to create a performance culture within the firm.

The second-most popular rationale for incentive pay among investors (79%/1.14) and third-most among directors (84%/1.16) was a response strongly contradicted by theory: “so that the CEO shares risks with investors and stakeholders, even if out of the CEO’s control.” While this implies inefficient risk-sharing, several respondents argued that it is unfair to insulate CEOs from a downturn. Free-text fields emphasized the importance of “shareholder alignment” and “to mirror shareholder experience.” One investor, in an interview, said that CEOs should be co-owners and be “there for the journey;” in a downturn, “it’s not fair that I have to take the pain and you don’t.” Thus, one reference point in Y , the vector that directors and investors use to assess fairness, may be a function of shareholder returns. While a board could argue “we won’t cut CEO pay because the recession isn’t her fault; providing this insurance allowed us to pay her less,” investors and stakeholders are unlikely to accept this argument – particularly since they do not see the contract that would have been offered without the insurance.

A different rationale for incentive pay is screening (e.g., Lazear, 2005), which we tested with “to attract/retain a high-ability or

³² Fehr et al. (1993) provide experimental evidence that agents reciprocate discretionary fair pay with discretionary effort.

Table 13

Q10: What would happen if you made the CEO's incentives more long-term? (-2=very unlikely outcome, 2=very likely outcome)

Investors: What would happen if companies made CEO incentives more long-term?

The first row of responses for each option (not italicized) is from directors (n=177), the second row (italicized) is from investors (n=135).

	Mean	-2	-1	0	1	2
The incentives would lose their effectiveness	0.20 ^{b,a}	6%	25%	25%	31%	13%
<i>We would have to pay the CEO more, which would outweigh any benefits</i>	<i>-0.98^{a,a}</i>	<i>30%</i>	<i>44%</i>	<i>21%</i>	<i>4%</i>	<i>1%</i>
<i>The CEO would need to be paid more, which would outweigh any benefits</i>	<i>0.14^{c,a}</i>	<i>7%</i>	<i>18%</i>	<i>37%</i>	<i>29%</i>	<i>9%</i>
We would be unable to attract/retain the CEO we want	-0.81 ^{a,a}	19%	51%	24%	4%	2%
<i>The board would be unable to attract/retain the CEO it wants</i>	<i>0.13^{-a}</i>	<i>7%</i>	<i>19%</i>	<i>34%</i>	<i>31%</i>	<i>8%</i>
The CEO would make better decisions	-0.95 ^{a,a}	28%	45%	21%	4%	1%
	-0.40 ^{a,a}	21%	25%	32%	15%	7%
	1.14 ^{a,a}	1%	4%	16%	34%	44%

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively.

Table 14

Q11a: Do you filter out industry conditions from all performance measures (e.g. by benchmarking against peers)? (Y/N)

Investors: Do you believe that industry conditions should be filtered out from all performance measures (e.g. by benchmarking against peers)?

Q11b: Why don't you filter out industry conditions from all performance measures? (-2=strongly disagree, 2=strongly agree)

Investors: Why don't you believe that industry conditions should be filtered out from all performance measures?

The first row of responses for each option (not italicized) is from directors (n=175/107), the second row (italicized) is from investors (n=135/98).

Panel A						
		Yes		No		
Do you filter out industry conditions from all performance measures?		37%		63%		
<i>Do you believe that industry conditions should be filtered out from all performance measures?</i>		<i>25%</i>		<i>75%</i>		
Panel B						
	Mean	-2	-1	0	1	2
The CEO should benefit from an industry upswing, since investors and stakeholders do	0.44 ^{a,a}	2%	9%	46%	29%	14%
<i>CEOs should benefit from an industry upswing, since investors and stakeholders do</i>	<i>0.05^{-a}</i>	<i>8%</i>	<i>19%</i>	<i>40%</i>	<i>24%</i>	<i>8%</i>
It is too difficult to define an appropriate peer group	0.20 ⁻	12%	21%	23%	21%	22%
<i>It is too difficult to define an appropriate peer group</i>	<i>-0.06⁻</i>	<i>9%</i>	<i>26%</i>	<i>37%</i>	<i>19%</i>	<i>9%</i>
We don't have information on peer performance for some measures	0.17 ⁻	18%	14%	21%	27%	20%
<i>The board doesn't have information on peer performance for some measures</i>	<i>-0.12⁻</i>	<i>14%</i>	<i>26%</i>	<i>27%</i>	<i>26%</i>	<i>8%</i>
Investors don't want us to filter out industry conditions*	0.11 ⁻	8%	14%	46%	21%	10%
Benchmarking all performance measures would lead to the CEO mimicking peers	-0.23 ^{b,b}	12%	29%	32%	24%	3%
<i>Benchmarking all performance measures would lead to the CEO mimicking peers</i>	<i>0.14^{-b}</i>	<i>10%</i>	<i>19%</i>	<i>30%</i>	<i>28%</i>	<i>13%</i>
Relative performance measures are less motivating for the CEO	-0.26 ^{a,-}	11%	32%	33%	21%	4%
<i>Relative performance measures are less motivating for CEOs</i>	<i>-0.46^{a,-}</i>	<i>18%</i>	<i>36%</i>	<i>24%</i>	<i>16%</i>	<i>5%</i>
In an industry upswing, not benchmarking performance keeps our pay competitive with peers	-0.32 ^{a,-}	18%	22%	36%	22%	2%
<i>In an industry upswing, not benchmarking performance keeps the pay competitive with peers</i>	<i>-0.17^{c,-}</i>	<i>10%</i>	<i>21%</i>	<i>47%</i>	<i>18%</i>	<i>3%</i>
The CEO is responsible for choosing what industries our firm operates in	-1.07 ^{a,a}	53%	21%	10%	8%	7%
<i>The CEO is responsible for choosing what industries our firm operates in</i>	<i>-0.42^{a,a}</i>	<i>18%</i>	<i>33%</i>	<i>29%</i>	<i>13%</i>	<i>7%</i>

The first superscript reports whether the mean is significantly different from zero, the second whether they are significantly different for directors and investors. a, b, and c represent statistical significance at the 1%, 5%, and 10% level, respectively. *Only directors were offered this response.

hard-working CEO." This also received strong support (87%/1.19 from directors and 69%/0.85 from investors). We also offered a response of "to motivate the CEO to improve outcomes other than long-term shareholder value." Interestingly, it received only modest support (52%/0.46 from directors and 53%/0.47 from investors), despite environmental, social, and governance ("ESG")-linked pay becoming increasingly frequent. Respondents may believe that alignment with long-term shareholder value already causes the CEO to take stakeholders into account, or that intrinsic incentives are sufficient. One investor wrote "What kind of outcomes, other than long-term shareholder value? Do CEOs really need incentives to 'do the right thing?'" An interviewee said that ESG targets would backfire as they would cause the CEO to improve only the ESG dimensions in the contract.

Directors (49%/0.37) assigned significantly greater importance than investors (15%/-0.45) "to match[ing] peer firm practice." This highlights an interesting paradox – recall from Table 5 that directors report that investors and their advisors (proxy agencies) force them to follow market practice more than they would like. Our results here, however, suggest that many investors appreciate heterogeneity – thus, directors

may have more latitude to tailor the structure of pay than they think.

5.3. The determinants of variable pay

To test theory predictions for the strength of incentives, Table 12 asked "What determines the split between variable and fixed pay?" Unexpectedly, the lowest responses were "CEO personal risk appetite" (22%/-0.46 for directors and 20%/-0.45 for investors) and "how risky our firm is" (16%/-0.44 and 47%/0.29).³³ These are key drivers of incentives in classic models such as Holmstrom and Milgrom (1987) – the higher firm risk and CEO risk aversion, the greater the compensating differential the CEO requires for variable pay. However, if the motivation for rewarding good performance is recognition and fairness, this trade-off is moot; instead, pay variability is driven by what is perceived to be the fair reward for performance. This is consistent with the mixed

³³ Beta testing identified "risk appetite" as the best way to translate "risk aversion" into practitioner language.

evidence for the link between risk and incentives summarized in [Prendergast \(2002\)](#) and [Edmans et al. \(2017\)](#); the low importance of risk aversion fits its weak relationship with incentives documented by [Becker \(2006\)](#).³⁴

The most popular response was “how much the CEO can affect firm performance” (62%/0.66 for directors and 75%/0.98 for investors). While consistent with standard principal-agent models, their mechanism is that, if effort is less effective, the principal induces less of it as it is not worth the risk premium required to do so; thus, incentives fall. This is inconsistent with the low support for the relevance of risk, and also the evidence in [Table 2](#) that incentives are much more important than the cost of pay. Interviews and text fields instead pointed to a different mechanism – if the CEO has a greater effect on performance, it is fair to reward her more for good performance.

The next two most popular responses from directors are not predicted by standard models: “investor or proxy advisor expectations” (60%/0.55) and “the split between fixed and variable pay in peer firms” (49%/0.37). Thus, many directors do not set incentives from first principles but follow market practice. Free-text responses included “The benchmark seemed to be set at roughly a third for each of salary, bonus & shares. ‘This is what everyone else does’” and “RemCos are being increasingly forced to ‘fit’ genuine motivational adjustments to pay into a rigid framework [...] governed by superficial numerical comparisons with other companies.” Again, this is surprising given that investors claim not to consider peer firm practices as important ([Table 11](#)).

5.4. Long-term incentives

Classic theories of managerial myopia (e.g., [Stein, 1988](#)) highlight the erosion of shareholder value that arises if CEO pay is tied to the short-term stock price. A natural solution is to pay the CEO according to long-term performance ([Edmans et al., 2012](#)). Even if the CEO demands a compensating differential for the greater risk, this may be outweighed by the benefits of superior decisions. However, horizons are short in practice ([Gopalan et al., 2014](#)) and lead to CEOs taking actions to boost short-term profits and stock prices ([Edmans et al., 2017](#); [Ladika and Sautner, 2020](#)), often at the expense of firm value ([Edmans et al., 2022](#)). [Table 13](#) studies the reasons for this apparent disconnect between theory and practice, asking “What would happen if you made the CEO’s incentives more long-term?”

By far the most popular response from investors was “the CEO would make better decisions” (78%/1.14). The three responses that suggest negative consequences – “the incentives would lose their effectiveness”, “we would have to pay the CEO more, which would outweigh any benefits”, and “we would be unable to attract/retain the CEO we want” – all scored below -0.8, with at most 6% agreeing with each.³⁵ As one fund manager wrote, “This would be a win win win win win. It would weed out CEOs that are in it for a quick buck, it would focus on long-term outcomes, and it would align CEOs with shareholders. If I could have a single bullet to improve corporate governance, this would be it.” Another said “We would get better alignment between CEO and owners. It is ridiculous that industries with a 5, 10, 15 year business/product cycle have a 1 and 3 year incentive program.”

Directors strongly disagree. “The CEO would make better decisions” obtained the weakest response (22%/-0.40), and 63% of directors agreed with at least one of the concerns. In the free text fields, directors explained that incentives are already sufficiently long-term and that further lengthening would reduce their effectiveness. Others argued that

shareholders are short-termist and would object. This discrepancy between investors’ stated preferences and directors’ views of those preferences is interesting and echoes the mismatch on investors’ preferences for tailoring.

5.5. Relative performance evaluation

[Holmstrom \(1982\)](#) showed that pay-for-performance contracts should benchmark performance against peers, to filter out fluctuations caused by external factors. This reduces the CEO’s risk, lowering the cost of compensation. In their critique of CEO pay, [Bebchuk and Fried \(2004\)](#) highlighted the lack of relative performance evaluation (“RPE”) as prime evidence for rent extraction. While recent studies show that the use of RPE is greater than in the past (see the survey by [Edmans et al. \(2017\)](#)), it is far from universal.

[Table 14](#), Panel A, therefore asked “Do you filter out industry conditions from all performance measures (e.g. by benchmarking against peers)?” 63% of directors and 75% of investors responded “No.” To those respondents, Panel B asked why they do not universally benchmark. The most popular answer among directors (43%/0.44, versus 33%/0.05 from investors) was that “the CEO should benefit from an industry upswing, since investors and stakeholders do.” This contradicts efficient risk-sharing, but is consistent with notions of fairness in two ways. First, if investors enjoy windfalls from industry conditions, it may be seen as fair for the CEO to gain also. Second, not benchmarking on the downside (see [Table 11](#)) means it may be seen as unfair to do so on the upside. If pay-for-luck is explained by fairness concerns, then it should be symmetric on both the upside and downside, and it should be equally prevalent in well governed firms. Both results are documented by [Daniel et al. \(2020\)](#).

The next two highest responses from directors were “It is too difficult to define an appropriate peer group” (43%/0.20, 29%/-0.06 for investors) and “We don’t have information on peer performance for some measures” (47%/0.17, 34%/-0.12). These are considerations absent from all models, which assume that a peer group exists and its performance is observable. One director explained that “for a large firm, there are few if any comparators which are similar enough.” On the observability of peer performance, one director wrote that “particularly non-financial measures are hard to benchmark.”

Investors responded most positively to “benchmarking all performance measures would lead to the CEO mimicking peers” (41%/0.14, versus only 27%/-0.23 for directors). The model of [Zwiebel \(1995\)](#) shows that benchmarking can cause the CEO to take insufficient good risk, but the free-text responses did not raise this concern. Instead, they pointed out that mimicry could lead to firms taking excessive risk (such as over-leveraging if peers are over-leveraging) or short-termist actions (cutting investment to boost margins if peers are doing so).

We also explored other theories for the lack of RPE – “Relative performance measures are less motivating for the CEO” ([Dittmann et al. 2013](#)), “In an industry upswing, not benchmarking performance keeps our pay competitive with peers” ([Oyer, 2004](#)), and “The CEO is responsible for choosing which industries our firm operates in” ([Gopalan et al. 2010](#)) – but these theories received little support.

There are more complex theories of pay-for-luck that we did not test because of the “as if” critique – they are so intricate that respondents may be unaware they are a driver. It may be that these alternative theories do explain pay-for-luck in reality. However, applying Occam’s razor, a more important reason is likely to be the inability to define an appropriate peer group or observe performance, or that pay-for-luck is not a concern to begin with.

5.6. Summary

Our questions on the structure of pay yield the following conclusions:

³⁴ Risk aversion may matter for other aspects of pay not studied by our survey, such as its convexity (see [Coles et al. \(2006\)](#)).

³⁵ One potential reason for the low response to the last option is that more long-term incentives may have opposite effects on attraction and retention. They may deter a CEO from joining, but once a CEO has joined they may encourage retention if forfeited upon departure ([Jochem et al., 2018](#)).

1. Directors and investors consider intrinsic motivation and personal reputation to be the most important sources of incentives for CEOs.
2. The primary reason for variable pay is to motivate the CEO to improve long-term shareholder value, but the mechanism is not necessarily that the CEO obtains utility from consuming the additional pay from good performance. Instead, variable pay provides ex-post recognition of performance, addressing the CEO's fairness concerns and preserving her intrinsic motivation.
3. The main determinant of pay variability is how much the CEO can affect firm performance. Firm risk and CEO risk aversion are not important. Directors feel constrained by peer firm practice and investor and proxy advisor expectations, hindering them from tailoring pay – even though investors do not consider following peer practice as important.
4. Investors believe that lengthening the horizon of CEO incentives would improve decisions, with few adverse consequences. Directors disagree. Some are concerned about the attraction/retention effects of such a change; others believe that incentives would become less effective.
5. Most directors and in particular investors do not believe that benchmarking of CEO performance measures should be universal. One reason is that they view it as fair for CEO pay to mirror the shareholder experience. A second is that, for many companies, it is difficult to define an appropriate peer group or obtain information on peer performance.

6. Generalizability

This section discusses the generalizability of our results, given the UK context and the survey being conducted during the COVID-19 pandemic. We discuss these two concerns in turn.

6.1. Generalizability beyond the UK

A natural question is whether our findings apply to other countries with different pay practices and regulations, despite widespread convergence. Since many of our most interesting results are unrelated to specific UK regulations, we expect them to carry over. For example, boards feeling constrained by external forces in their ability to optimize CEO pay, boards and investors disagreeing about the consequences of reducing CEO pay, and CEOs wanting to be paid fairly are likely to be universal. However, the finer details might differ, so repeating similar surveys in other countries would be useful.

As discussed in Section 2.3, the UK has several attractions in terms of generalizability. For example, both UK pay levels and structures lie between the US and Europe; in contrast, a US study would be less likely to generalize to Europe, and vice-versa. However, for researchers interested in the US, differences with UK practices may affect the applicability of some findings.

First, without adjusting for firm size and risk, the level of pay is significantly lower in the UK, but even after adjusting some gap remains. For example, in 2021, median “single figure” pay for FTSE 100 firms was £3.6 million (approximately \$4 million), compared to \$14.4 million in the S&P 500. The CEO-to-worker pay ratio, which many investors and stakeholders focus on, is 81:1 in the UK versus 193:1 in the US.³⁶ This suggests that downward pressure on the level of pay may be stronger in the UK. Our first finding (Table 2) was that both directors and investors believe that lowering pay is significantly less important than retaining and motivating the CEO. If anything, this result should be even stronger in the US, given that directors and investors have allowed pay to rise to higher levels.

Moreover, higher levels of pay do not mean that CEO pay is

unconstrained in the US, so our results on the importance of external forces should also be relevant. For example, mean (median) pay in the S&P 500 fell by 40% (7%) between 2000 and 2014 (Edmans et al. 2017). In addition, prior literature has documented the influence of external parties on pay in the US (e.g., Malenko and Shen (2016) for proxy advisors). More recently, a 50% increase in say-on-pay vote defeats in the S&P 500 in 2021 was attributed to COVID-related pay actions, suggesting that alignment with the experience of employees and shareholders is also important in the US.³⁷ However, while pay is constrained in the US, the relative importance of the various constraints may differ from the UK. Differences in pay levels also mean that our results on whether the level of CEO pay is too low, too high, or about right may not generalize, so we relegated them to Online Appendix A.

A second difference is in the structure of pay, with options being used much more in the US. We thus did not ask questions on the mix of stock and options, and our questions on variable pay (Tables 11 and 12) did not specify whether it comes from stock, options, LTIPs, or bonuses. We also did not ask questions on features of pay that may be less prevalent in the UK, such as inside debt (defined benefit pensions and deferred compensation), or significantly lower in the UK, such as severance pay.

A third difference is that institutional ownership is higher and more concentrated in the US. Institutional investors on average held more than 80% of the equity of S&P 500 firms in 2019, compared to 62% for the FTSE 100 (which still gave the UK the second highest institutional ownership in the OECD).³⁸ The “big three” institutional investors in the US (Vanguard, State Street, and BlackRock) held 21% of the equity of the average S&P 500 firm as of 2017 (Bebchuk and Hirst, 2019), versus only 9% for the three largest FTSE 100 investors (BlackRock, Vanguard, and Norges Bank).³⁹ This may limit the generalizability of our finding that investors feel they have too little influence on CEO pay, so we relegated it to Online Appendix A.

Our other questions asked about investors' objectives and the factors they consider, not their influence. It might be that a larger stake provides investors with greater incentives, thus inducing them to consider more firm-specific factors. However, Tables 11 and 12 show that UK investors would prefer more tailoring, suggesting that their smaller stakes do not hinder firm-specific analysis. In addition, Edmans and Holderness (2017) show that, if investors have capacity constraints, an investor's incentive to monitor a firm depends on its stake relative to its overall portfolio, rather than to firm value. Because the “big three” own large stakes in many firms, they are spread thinly. Thus, their ability to monitor any particular firm may not be markedly different from investors in the UK.

Finally, some regulations differ between the US and the UK. Because of UK investors' binding policy vote, in Tables 4 and 5 we studied the effects of “restrictions from our existing approved pay policy” on CEO pay. This has no implications for the US since, unlike in the UK or EU, there is no binding policy vote. Table 5 also showed that only 36% of UK directors reported that restrictions from regulation or governance codes caused them to offer an inferior structure of pay. Historically, this figure would likely have been higher in the US. Differences in accounting treatment favored options over shares, and tax rules, specifically performance-based pay being excluded from the \$1 million tax-deductibility limit, favored variable pay. However, options have been expensed in the US since 2006, and all pay above \$1 million has been non-deductible since 2018.

6.2. Influence of the pandemic

Another potential concern is that the survey was launched during

³⁷ See Mercer: “Record number of say-on-pay failures in 2021.”

³⁸ See Greenspoon (2019) for the US, FactSet: “Institutional Ownership in the UK” for the UK, and De La Cruz et al. (2019) for the OECD.

³⁹ See FactSet: “Institutional Ownership in the UK.”

³⁶ See Deloitte: “Directors' remuneration in FTSE 100 companies” for the UK and FW Cook: “CEO Pay Ratio Among S&P 500 Companies” for the US.

COVID-19. Section 2.3 described advantages of this timing, but a disadvantage is that the responses may be influenced by the pandemic. Even though we asked about the objectives, constraints, and determinants of pay in general, the pandemic may have made certain factors more salient. For example, respondents may have felt that they are constrained in setting pay even though, in normal times, such constraints are less binding.

However, external pressures on executive pay have been common in the UK and many other countries, not just during the pandemic. Already in 1995, the influential Greenbury Report, commissioned by the Confederation of British Industry, made several recommendations, most of which were incorporated into the UK's Corporate Governance Code in 1998. In 2002, the UK government made these rules statutory, further enhancing reporting requirements and introducing say-on-pay. The financial crisis of 2008 caused direct regulation of bankers' pay and public pressure to reduce pay in a recession. In 2010, the UK's new Stewardship Code for asset managers obligated its signatories to monitor actively, and since 2012 it lists remuneration as one of the areas of engagement.

In the "Shareholder Spring" of 2012, there were protest votes by institutional investors against executive pay at several large UK companies, and in 2013, the government again enhanced pay disclosure and introduced a binding second shareholder vote, citing "unsustainable ratcheting-up of executive pay." In 2015, outrage over WPP CEO Martin Sorrell's £70 million pay package led to the media and policymakers scrutinizing CEO pay across all companies and putting pressure on directors and investors, and every year the High Pay Centre, a think tank, releases a widely publicized report protesting high CEO pay. Thus, while the pandemic may have reinforced external constraints, it is not clear its effect was different from other economic downturns. However, just like the generalizability to other countries, we cannot formally show this. For future research, it would be fruitful to conduct similar surveys in other countries and different economic conditions.

7. Conclusion

This paper surveyed directors and investors on the objectives, constraints, and determinants of CEO pay. Our results show that many standard assumptions of executive pay models do not describe how pay is actually set, and they suggest alternatives that bring the models closer to reality.

Boards face a much larger set of constraints than participation and incentives, frequently causing them to offer lower pay and more one-size-fits-all structures. The strongest constraint is the need to obtain shareholder approval, suggesting that directors and investors disagree on how to maximize shareholder value. Investors perceive the need to avoid controversy with employees, customers, and policymakers as important additional constraints. Even for the standard participation and incentive constraints, investors and directors disagree on their importance. Investors want boards to lower the level of pay and align incentives more to long-term shareholder value. In contrast, directors believe that investors underestimate the difficulties of attracting and retaining CEOs, and that implementing investors' wishes would demotivate the CEO or precipitate her departure.

Turning to the determinants of pay, fairness concerns play an important role in both the level and structure of CEO pay. Starting with the level of pay, investors and especially directors believe that the CEO needs to be paid competitive amounts, even absent recruitment or retention concerns, because the failure to do so would be viewed as unfair and undermine the CEO's intrinsic motivation. CEOs are believed to assess their pay against their expectation of a fair reward, rather than based only on the consumption utility it provides.

Moving to the structure of pay, both directors and investors believe that pay should be linked to performance, even though they view financial incentives as less important motivators than intrinsic motivation and the CEO's personal reputation. Financial incentives are

believed to interact with these other drivers – not rewarding the CEO for good performance would be seen as unfair and undermine her motivation, but visibly doing so reinforces it. Thus, incentive pay provides ex-post recognition of performance, not just consumption incentives. Pay incentives, by delivering greater recognition, play a special role over and above portfolio incentives, whereas standard models view them as fungible.

The survey responses and interviews identified two other fairness considerations that explain the link between pay and performance. One is that CEOs are expected to share external shocks with investors and stakeholders, in contrast to optimal risk-sharing. A second is that, if employees' pay is linked to their performance, it is fair for the CEO's pay to be similarly sensitive.

Our results point towards a more complex but also more interesting model of CEO pay than currently used in the academic literature. Investors and especially directors believe that CEOs evaluate their compensation relative to a set of reference points, such as the pay of other CEOs, their past pay, and their perceived contribution to the firm. In turn, directors and investors evaluate CEO pay relative to a set of potentially different reference points, such as the pay of other CEOs, past CEO pay, the pay of employees and other top executives, and shareholder returns.

Our findings suggest a number of potential directions for future research. Starting with theory, the standard assumption of a single principal – a shareholder-aligned board – does not capture the complexity of the pay-setting process. The opposite assumption of the board maximizing pay, implicit in the "rent-extraction" view, is also unrealistic. Instead, pay is set by a board that seeks to maximize shareholder value, but also must obtain the approval of shareholders who have different information, beliefs, or objectives. Our results also call for models where pay is influenced by fairness concerns, felt by investors, directors, and CEOs, with a multitude of potential reference points. Chaigneau et al. (2023) is a first step.

Theorists may consider some of the practices we document here – such as following peer practice rather than designing contracts from first principles – as suboptimal and prefer to study how pay "should" be set. If so, they should acknowledge that their models are normative rather than positive. As a result, it may not be appropriate to evaluate them according to their empirical fit but instead the realism of their assumptions, which our survey may help guide.

The survey results also suggest that certain inefficient features of observed contracts may not be as puzzling as previously believed, so they may not need to be "explained" by future models. For example, if pay-for-luck is caused by fairness considerations and the difficulty of defining a peer group or observing its performance, then there is less need for models justifying pay-for-luck. Similarly, if variable pay is driven by the need to provide recognition rather than consumption incentives, the mixed evidence on the link between incentives and risk is less puzzling.

Future empirical analyses might further distinguish between the "weak/uninformed boards" and "uninformed investors" explanations for the disagreements between directors and investors. For example, researchers could study how deep the labor market for CEOs actually is, and what are the consequences (for CEO retention or performance) of changes to the level or structure of pay that make the contract less attractive. These consequences might differ according to whether these changes are made in one firm in isolation, or across the board. Another implication is that pay incentives and portfolio incentives may not be fungible and thus should be separated out in empirical analyses. Finally, it would be fruitful to study the importance of fairness in executive pay and how it is defined – i.e., what reference points CEOs, directors, and investors use to assess it.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jfineco.2023.103718](https://doi.org/10.1016/j.jfineco.2023.103718).

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